Prof. Lourdes F. Vega

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PROFILE AND SKILLS

- PhD in Physics with strong background in Engineering and 30+ years of professional experience in Research, Teaching, Innovation and Strategy, in academia and industry, in Europe, USA and the Middle-East.
- Internationally recognized for scientific and technical contributions in molecular thermodynamics and computational modeling applied to sustainable processes and products; focused on CO_2 capture and utilization, hydrogen, clean energy, catalytic and adsorbent materials, alternative solvents and water treatment.
- Demonstrated capacity for research leadership and strategic planning at a senior level.
- Strong communication, influence, organizing and planning skills – able to work easily across functions and with different teams and cultures, ranging from education and research to business and marketing. Able to quickly and efficiently adapt to different work environments and needs.

EDUCATION

Postdoctoral Associate, School of Chemical Engineering, Cornell University, Ithaca, NY (USA), 1992-1995
Visiting Scholar, Department of Chemical Engineering, University of Southern California, LA (USA), 1990-1992
Ph.D. in Physics, University of Sevilla (Spain), in collaboration with the Chemical Engineering Department, University of Southern California, Los Angeles, CA (USA), 1992.
M.S. in Physics (Licenciada con Grado). University of Sevilla, Spain, 1990.
B.S. in Physics (Licenciada). University of Sevilla, Spain, 1988.

PROFESSIONAL EXPERIENCE

2019- **Director, Research and Innovation Center on CO_2 and H_2, Khalifa University of Science and Technology-** Abu Dhabi, United Arab Emirates.
2016- **Joint Chair Professor, Gas Research Center, Khalifa University of Science and Technology-** Abu Dhabi, United Arab Emirates.
2015- **Founder and Chief Executive Officer (CEO), Alya Technology & Innovation S.L.** (Barcelona, Spain)**-** Start up company focused on R&D and innovation in sustainability from a broad perspective.
2012-15 Global Business Technology Manager, Air Products Group - Global R&D strategy and implementation of new and improved products in Agri-Food, Water and CO_2 applications. 35 gate processes projects.
2007-15 General Director, MATGAS – Research Center created as a Joint venture between the company Air Products (main owner), the National Research Council of Spain (CSIC) and the Autonomous University of Barcelona.
2007-15 R&D Director, Carburos Metálicos SA, Air Products Group. Member of the Executive team of Directors.
2003-11 Senior Research Scientist, CSIC. - Materials Science Institut of Barcelona (ICMAB). *Consejo Superior de Investigaciones Científicas* (National Research Council of Spain - CSIC).
2001-03 Deputy Director (Associate Dean) for International and External Affairs. - School of Chemical Engineering. Universitat Rovira i Virgili (URV). Tarragona. Spain.
1996- 99 President - PhD Program, Department of Chemical Engineering. URV, Tarragona, Spain. International program 70+ students in 1999.
1995-03 Associate/Professor of Chemical Engineering. Dept. of Chemical Engineering. URV. Tarragona, Spain.
1992-95 Post Doctoral Associate- School of Chemical Engineering. Cornell University, Ithaca, NY, USA.
1990-92 Visiting Scholar – Dept. of Chemical Engineering. U. of Southern California, Los Angeles, California, USA.

AWARDS AND RECOGNITIONS

2020 Mohammed Bin Rashib Medal to the Scientific Distinguishment – for her contributions in clean energy and sustainable products.
2019 Visa of Talent - recipient of one of the 20 first long-term visas granted by the UAE Cabinet to grow the country's status as an incubator of talent and innovation.
2018 Finalist for the **Mohammed bin Rashid Medal for Scientific Excellence** (7 finalists out of 200 candidates), UAE
2017 Elected Member, Academy of Mathematics, Physical Chemistry and Natural Sciences of Granada (Spain).

2017 Finalist for the **Academic of Distinction Award, LEWA** (Leadership Excellence for Women Award recognizing professional excellence in women from the Oil & Gas Industry).

2014 Award by the Catalan Foundation for Research and the Generalitat of Catalonia to the **Public-Private Partnership for R&D and Innovation** (to MATGAS).

2013 Award of the Fundacion BBVA and Spanish Royal Society of Physics, in **Physics, Innovation and Technology,** granted by "*the outstanding and original manner that she managed to bring physics to the industrial world*".

2010 Physicist of Excellence, Recognition from the Collegiate of Physicists of Spain in 2010, highlighting the work of physicists who have reached a **level of excellence in their profession**.

Chosen as one of the **16 Outstanding Women Scientists** in an exhibition of Women in Science, Catalonian Association of Scientific Communication, 2010.

Scientific curator of the **Smart Future Minds Award**, as part of the "smart urban stage" exhibition held in Barcelona (April-June, 2011) for the sustainable, future-oriented ideas, organized by Mercedes Benz in European Cities.

Teaching awards

Students 1st place award in Energy and Environment, UAE Think Science competition (2018), students' project co-advisor (with Dr. Nahla Alamoodi).

Students 2nd place award in Energy and Environment, UAE Think Science competition (2017), students' project co-advisor (with Dr. Maryam Khaleel).

Award to the **Most Valuable Teacher in the PhD program**, granted by the graduate students of the program. Department of Chemical Engineering, URV. Academic year 2002-03.

Two teaching awards as a member of the academic staff of the School of Chemical Engineering, URV. (2001 Vicenç Vives, Generalitat de Catalunya; 2002 Ministerio de Educación y Ciencia). Granted for the innovative integrated approach for teaching the undergraduate curricula, through integrated projects.

EXTERNAL AFFILIATIONS

2019- **Editor-in-Chief,** Journal of Molecular Liquids of Elsevier (Q1, impact factor 4.561). Since October

2019- **Member of the International Scientific Advisory Board of IREC,** research foundation for energy in Catalonia, Spain.

2016- **Independent Member of the Board of Directors, ERCROS S.A.,** Spain.

2016- **Invited Professor,** University of Zaragoza, Spain

2015- **Member of the International Industrial Advisory Board of the Center for Nature Inspired Engineering (CNIE),** Chemical Engineering Department, University College of London, UK.

Member of the Editorial Advisory Board of the Journal of Supercritical Fluids, Fluid Phase Equilibria, Industrial and Engineering Chemistry Research and the International Journal of Thermophysics

Member of the Americal Institute of Chemical Engineers (AIChE)

Member of the Americal Chemical Society (ACS)

Member of the Collegiate of Physicists of Spain

Member of the Royal Society of Physics and Chemistry of Spain

Member of the Spanish Technological Platform of Hydrogen and Fuel Cells

Independent expert evaluator of R&D&I Projects for different funding institutions: European Projects: FP6, FP7 and H2020 program; USA projects (NSF, DOE); French Projects (ANR), Spanish projects and Autonomous Projects.

International experience in **event planning and execution** (10 technical days, 10 international/national) conferences.

Member of the **scientific committee** of 7 international conferences and 3 national conferences.

Chairperson of the international organizing committee for the PPEPPD 2022 meeting, an international conference devoted to phase equilibria and thermophysical properties for process and product design, held every 3 years, the next edition to be held in Europe.

Member of the **Technical Programme Committee** (TPC) of the Greenhouse Gas Control Technologies (GHGT) conference series for the GHGT-15, to be held in Abu Dhabi in 2020.

SENIOR MANAGEMENT EXPERIENCE (REVERSE CHRONOLOGICAL ORDER)

Director of the Research and Innovation Center on CO_2 and H_2, KU (2019-). A center in the area of clean and future energy, devoted to CO_2 capture, CO_2 utilization and H_2 production, storage and distribution, Integrated by 15 Faculty from different colleges and 40+ researchers.

Mentor to Junior Faculty: acting as the Senior Mentor for Assistant Professors at Khalifa University (2016-)

Member of the University Promotion Review Committee Appointments (UPRC), KU (2017-)**.** Committee advising the President on the internal promotions and external faculty hiring.

General Director of MATGAS 2000 AIE. Defined and executed **triannual strategic plans** for MATGAS: mission, vision, strategic goals, milestones and deliverables, budget. Full accountability for the center (including hiring, personnel, laboratories, projects, finances and safety). Acting as link with the 3 partners (Air Products, Universitat Autonoma de Barcelona and CSIC) – understanding their strategy, building common goals.

Global Business Technology Manager: responsibility for the **technology, strategy and applications development** for the Agri-Food, Water Treatment and CO_2 commercial segments for Air Products. Leading a multidisciplinary team of 40 direct report technical people, coordinated with the marketing and commercial teams in Europe, America and Asia. Connection with external organizations in R&D, in academia and companies.

Research & Development Director of Carburos Metálicos - direction and management of new product development, Research labs for the company and connection with Universities and other companies for Project development and innovation. Member of the Executive Board of Directors in Spain.

Member of the International Industrial Advisory Board of the Center for Nature Inspired Engineering (CNIE) at the University College of London, UK (2015-). Providing advice to the Director and follow up on the strategy and programs to succeed in research, development and deployment of developed new products and procedures.

Technical **lead and coordinator** of three large public-private consortia: the CENIT SOST-CO2 project (2008-2011), 14 companies and 31 research centers, budget: 26.4 MM €; the EU project H_2TRUST (2013-2015), with 9 partners of 5 EU countries, budget: 1.2MM €; and the NUCLI BioQuim_rescue (2013-2015), a Catalan/Feder funded project, focused on water treatment. 4 companies, 3 research institutions. Budget: 770000€.

Leader of the Working Group CO_2 utilization of the Spanish Technological Platform for CO_2 (2008-2015)

Deputy Director (Associate Dean) for International Relations and Industry, School of Chemical Engineering, Universitat Roviria I Virgili, Tarragona, Spain (2001-2003). Link between the School of Chemical Engineering, Companies, Secondary Schools and International Universities. Strategy and implementation.

First President (Dean) and Promoter of the International PhD Program in Chemical Engineering, Universitat Rovira I Virgili, 70 students from 20 Countries (1996-99).

PUBLICATIONS (SEE ENCLOSED LIST)

5 patents, 200+ publications in referred journals, 2 books as a sole author, 1 book and 6 monographs as co-author. 50+ book chapters.

Citations: (8 July 2020): Scopus: 5668 citations, h-index: 41

Google Scholar: 7105 (3393 since 2015). H-index: 47 (31 since 2015); I10 index= 129 (98 since 2015)

600+ presentations in international conferences. ~70 invited/plenary and 150+ additional presentations since 2011.

Supervised 11 PhDs (4 under execution), 26 MSc (6 under execution), 23 Postdoctoral Associates/Research Scientists.

PROJECTS (SEE ENCLOSED LIST OF SELECTED RECENT PROJECTS)

Areas of expertise: molecular modelling and simulations, clean energy, CO_2 capture and utilization, oil and gas, hydrogen, nanotechnology, alternative solvents, water treatment, sustainability.

Director of the Research and Innovation Center on CO2 and H2, with 8 projects under execution related to materials for CO2 capture, CO2 utilization, H2 production, storage and distribution.

Leader of 3 multiannual large public-private consortia. Total budget directly managed ~ 30MM€ (valued ~100MM€)

Participation as a scientific lead in 50+ competitive research grants (EU, USA, Spain, Middle East) and 60 R&D industrial projects (EU, USA, Middle East).

Participation in 13 EU funded projects (FP6, FP7, H2020), coordinating 1 of them.

TEACHING EXPERIENCE (SEE ENCLOSED LIST)

25 different courses, at the undergraduate and graduate level, in **5 Universities** (EU, USA and ME): Petroleum Institute, now part of the Khalifa University of Science and Technology (UAE), Universitat Autònoma de Barcelona (Spain), Universitat Rovira i Virgili (Spain), Cornell University (USA) & Universidad de Sevilla (Spain).

Invited Visiting Professor in the US, EU, Chile.

LANGUAGES

Fluent in Spanish, English and Catalan. Knowledge of French, Portuguese and Chinese.

PUBLICATIONS AND SUPERVISED WORK

TOTAL NUMBER OF CITATIONS
Citations: (8 July 2020): Scopus: 5668 citations, h-index: 41
Google Scholar: 7105 (3393 since 2015). H-index: 47 (31 since 2015); I10 index= 129 (98 since 2015)
200+ papers in international journals (ISI/SCR or external referees). **2 books as a sole author**. 1 book and 6 monographs as co-author; 50+ book chapters.
5 international **patents**.
11 PhD Thesis as supervisor (4 under development); **26 Master Thesis** as supervisor (6 under development).
23 supervised **Postdoctoral Associate**

PUBLICATIONS INCLUDED IN THE JOURNAL OF SCIENCE CITATION INDEX

1 **Vega, L.**, de Miguel, E., Rull, L.F., Jackson, G., McLure, I.A.
Phase equilibria and critical behavior of the square-well fluids of variable width by Gibbs ensemble Monte Carlo simulation.
Journal of Chemical Physics, 96, 2296-2305 (1992)
doi.org/10.1063/1.462080

2 **Vega, L.F.**, Shing, K.S., Rull, L.F.
A new algorithm for molecular dynamics simulations in the grand canonical ensemble
Molecular Physics, 82, 439-453 (1994).

3 **Vega, L.F.**, Panagiotopoulos, A.Z., Gubbins, K.E.
Chemical potentials and adsorption isotherms of polymers confined between parallel plates.
Chemical Engineering Science, 49, 2921-2929 (1994).

4 Müller, E.A., **Vega, L.F.**, Gubbins, K.E.
Theory and simulation of associating fluids: Lennard-Jones chains with association sites.
Molecular Physics, 83 (6), 1209-1222 (1994).

5 Müller, E.A., **Vega, L.F.**, Gubbins, K.E.
Molecular simulation and theory of associating chain molecules
International Journal of Thermophysics, 16 (3), 705-713 (1995)

6 Tsang, P.C., White, O.N., Perigard, B.Y., **Vega, L.F.**, Panagiotopoulos, A.Z.
Phase equilibria in ternary Lennard-Jones mixtures.
Fluid Phase Equilibria, 107, 31-44, (1995)

7 **Vega, L.F.**, Panagiotopoulos, A.Z., Gubbins, K.E.
Chemical potentials and adsorption isotherms of polymers confined between parallel plates.
Chemical Engineering Science, 50, 2505 (1995). Correction/addition.

8 Müller, E.A., **Vega, L.F.**, Gubbins, K.E., Rull., L.F.
Adsorption isotherms of associating chain molecules from Monte Carlo simulations.
Molecular Physics, 85, 9-22 (1995).

9 **Vega, L.F.**, Müller, E.A., Rull, L.F., Gubbins, K.E.
Mixtures of associating and non-associating chains on activated surfaces: a Monte Carlo approach.
Molecular Simulation, 15,141-154 (1995).

10 **Vega, L.F.**, Müller, E.A., Rull, L.F., Gubbins, K.E.

Effect of surface active sites on the adsorption of associating chains on activated surfaces: A Monte Carlo approach.
Adsorption, 2, 59-68 (1996).

11 Müller, E.A., Rull, L.F.,**Vega, L.F.**,Gubbins, K.E.
Adsorption of water on activated carbons: A molecular simulation study.
Journal of Physical Chemistry, 100, 1189-1196 (1996).
doi.org/10.1021/jp952233w

12 Vega, L.F., Muller, E.A.; Rull, L.F.; Gubbins, K.E.
Adsorption isotherms of associating fluids in slit-like pores. A Monte Carlo Study
Fundamentals of Adsorption, Kluwer Academic Publisher, ed. M.D. Le Van , 993-1000, (1996)

13 Blas, F.J.; **Vega, L.F.**
Thermodynamic behavior of homonuclear and heteronuclear Lennard-Jones chains with association sites from theory and simulation.
Molecular Physics, 92, 135-150 (1997).
doi.org/10.1080/002689797170707

14 Blas, F.J.; **Vega, L.F.**
Prediction of binary and ternary diagrams using the SAFT equation of state.
Industrial and Engineering Chemistry Research, 37, 660-674 (1998)

15 Bandosz, T.J.; Blas, F.J.; Gubbins, K.E.; McCallum, C.L.; McGrother, S.C.; Sowers, S.L.; **Vega, L.F**.
Molecular Modeling of Selective Adsorption from Mixtures
Materials Research Society Symposium Proceedings, 497, 231- (1998), Materials Research Society, eds: N.M. Rodriguez, S.L. Soled, J. Hrbek

16 Blas, F.J.; **Vega, L.F.**; Gubbins, K.E
Modeling New Adsorbents for Ethylene/Ethane Separations by Adsorption via π-Complexation
Fluid Phase Equilibria, 150-151, 117-124 (1998)

17 Blas, F.J.; **Vega, L.F.**
Critical behavior and partial miscibility phenomena in binary mixtures of hydrocarbons by the Statistical Associating Fluid Theory
Journal of Chemical Physics, 109, 7405-7413 (1998).

18 Mackie, A.D.; Hernández-Cobos, J.; **Vega, L.F.**
Liquid vapor equilibria for an ab-initio Model for Water
Journal of Chemical Physics, 111, 2103-2109 (1999)

19 **Vega, L.F.**, Blas, F.J.
Tricritical phenomena in chain-like mixtures from a molecular-based equation of state
Fluid Phase Equilibria, 171, 91-104 (2000)

20 Mackie, A.D.; Hernández-Cobos, J.; **Vega, L.F.**
Predicting liquid-vapour equilibria for water using an ab-initio potential from histogram reweighting Monte Carlo simulations
Molecular Simulation, 24, 63-69 (2000)

21 Hernández-Cobos, J.; Mackie, A.D.; **Vega, L.F.**
The hydrophobic hydration of methane as a function of temperature from histogram reweighting

Monte Carlo simulations
Journal of Chemical Physics, 114, 7527-7535 (2001)

22 Pàmies, J.C.; **Vega, L.F.**
The vapor liquid equilibria and critical behavior of heavy n-alkanes using transferable parameters from the Soft-SAFT equation of state
Industrial and Engineering Chemistry Research, 40, 2532-2543 (2001)

23 Blas, F.J.; **Vega, L.F.**
Improved vapor-liquid predictions for Lennard-Jones chains from the Statistical Associating Fluid Dimer Theory: comparison with Monte Carlo simulations
Journal of Chemical Physics, 115, 4355-4358 (2001).

24 Blas, F.J.; **Vega, L.F.**
Thermodynamic properties and phase equilibria of branched chains using first and second order Wertheim's Thermodynamic Perturbation Theory
Journal of Chemical Physics, 115, 3906-3915 (2001)

25 Figueroa-Gerstenmaier, S.; **Vega, L.F**., Blas, F.J.; Gubbins, K.E.
Molecular Model of gamma-alumina: Nitrogen Adsorption and Pore Size Distribution
AIChE Symposium Series, 325, vol 97, 317-320 (2001). Ed: Foundations of Molecular Modeling and Simulations

26 Colina, C.M.; Turrens, L.F.; Gubbins, K.E.; Oliveras-Fuentes, C.; **Vega, L.F.**
Predictions of the Joule-Thomson inversion curves for the n-alkanes series and carbon dioxide from the Soft-SAFT equation of state
Industrial and Engineering Chemistry Research, 41,1069-1075 (2002)

27 Pàmies, J.C.; **Vega, L.F.**
Critical properties of homopolymer fluids studied by a Lennard-Jones Statistical Associating Fluid Theory
Molecular Physics, 100 (15), 2519-2529 (2002)

28 Hernández-Cobos, J.; **Vega, L.F.**
The influence of the density in the hydrophobic hydration of methane from Monte Carlo simulations
Journal of Molecular Liquids, 101, 113-125 (2002)

29 Figueroa-Gerstenmaier, S.; Blas, F.J.; Bonet Avalos, J.; **Vega, L.F.**
Application of the Fundamental Measure Density Functional Theory to the Adsorption in Cylindrical Pores
Journal of Chemical Physics, 118, 830-842 (2003)

30 Pàmies, J.C.; McCabe, C.; Cummings, P.T.; **Vega, L.F.**
Coexistence Densities of Propane by Canonical Molecular Dynamics and Gibbs Ensemble Monte Carlo Simulations
Molecular Simulation, 29 (6-7), 463-470 (2003)

31 Florusse, L.J.; Pàmies, J.C.; **Vega, L.F**.; Peters, C.J.; Meijer, H.
Solubility of hydrogen in heavy n-alkanes: experiments and SAFT modeling
AIChE Journal, 49, 3260-3269 (2003)

32 Figueroa-Gerstenmaier, S.; Bonet Avalos, J.; Gelb, L.D.; Gubbins, K.E.; **Vega, L.F.**

Pore Size Distribution of Porous Glasses: a Test of the Independent Pore Model
Langmuir, 19, 8592-8604 (2003)

33 Dias, A.M.A.; Pàmies, J.C. ; Coutinho, J.A.P.; Marrucho, I.M.; **Vega, L.F.**
SAFT modelling of the Solubity of Gases in Perfluoroalkanes
Journal of Physical Chemistry B, 108, 1450-1457 (2004)

34 Herdes, C.; Pàmies, J.C. ; Marcos, R.M.; **Vega, L.F.**
Thermodynamic properties and aggregate formation of surfactant-like molecules from theory and simulation
Journal of Chemical Physics, 120, 9822-9830 (2004)

35 Dias, A.M.A.; Caço A-I.; Coutinho, J.A.P.; Santos L.B.; Piñeiro M.M.; **Vega, L.F**.; Costa Gomes M.F.; Marrucho I.M
Thermodynamic properties of perfluoro-n-octane
Fluid Phase Equilibria, 225, 39-47 (2004)

36 Duque, D.; **Vega, L.F**.
Some issues on the calculation of interfacial properties by molecular simulation
Journal of Chemical Physics, 121, 8611-8617 (2004)

37 Llovell, F.; Pàmies, J.C.; **Vega, L.F**.
Thermodynamic properties of Lennard-Jones chain molecules. Renormalization-group corrections to a Statistical Associated Fluid Theory (soft-SAFT)
Journal of Chemical Physics, 121, 10715-10724 (2004)

38 Duque, D.; Pàmies, J.C.; **Vega, L.F**.
Interfacial properties of Lennard-Jones chains by direct simulation and Density Gradient Theory
Journal of Chemical Physics, 121, 11395-11401 (2004)

39 Mejía, A; Segura, H.; **Vega, L.F**.; Wisniak, J.
Simultaneous predictions of interfacial tension and phase equilibria in binary mixtures. An approach based on cubos equations of state with improved mixing rules
Fluid Phase Equilibria, 227, 225-238 (2005)

40 Sánchez-Montero, M.J.; Herdes, C.; Salvador, F.; **Vega, L.F**.
New insights into the adsorption isotherm interpretation by a coupled molecular simulation-experimental procedure
Applied Surface Science, 252, 519-528 (2005)

41 Herdes, C.; Abelló, S.; Santos, M.A.; Medina, F.; **Vega, L.F**.
Search for a reliable methodology for PSD analysis based on a combined molecular simulation-experimental approach. The case of PHTS materials
Applied Surface Science, 252, 538-547 (2005)

42 Mejía, A.; Pàmies, J.C.; Duque, D.; Segura, H.; **Vega, L.F**.
Phase and interface behaviors in type-I and type-V Lennard-Jones mixtures: theory and simulations
Journal of Chemical Physics, 123, 034505 (2005)

43 Dias, A.M.A.; Gonçalves, C.M.B.; Caço, A.I.; Santos, L.M.N.B.F.; Piñeiro, M.M. ; **Vega, L.F**.; J.A.P. Coutinho, I.M. Marrucho
Densities and vapour pressures of highly fluorinated compounds

Journal of Chemical Engineering Data, 50, 1328-1333 (2005)

44 Pedrosa, N.; Pàmies, J.C.; Coutinho, J.A.P.; Marrucho, I.M.; **Vega, L.F**
Phase equilibria of ethylene glycol oligomers and their mixtures
Industrial and Engineering Chemistry Research, 44, 7027-7037 (2005)

45 Herdes, C.; Santos, M.A.; Medina, F.; **Vega, L.F**
Pore Size Distribution Analysis of Selected Hexagonal Mesoporous Silicas by Grand Canonical Monte Carlo simulations
Langmuir, 21, 8733-8742 (2005)

46 Hernandez-Cobos J, Saint-Martin H, Mackie AD, **Vega L.F**., Ortega-Blake I.
Water liquid-vapor equilibria predicted by refined ab initio derived potentials
Journal of Chemical Physics, 123, 044506 (2005)

47 Olivet, A.; Duque, D.; **Vega, L.F**.
Sulphur Hexafluoride's Liquid-Vapor Coexistence Curve, Interfacial Properties and Diffusion Coefficients as Predicted by a Simple Rigid Model
Journal of Chemical Physics, 123, 194508 (2005)

48 Dias, A.M.A.; Pàmies, J.C.; **Vega, L.F**.; Coutinho, J.A.P.; Marrucho, I.M.
Modelling the solubility of gases in saturated and substituted perfluoroalkanes
Polish Journal of Chemistry, 80, 143-152 (2006)

49 Llovell, F.; **Vega, L.F**.
Global Fluid Phase Equilibria and Critical Phenomena of Selected Mixtures using the Crossover soft-SAFT Equation
Journal of Physical Chemistry B, 110, 1350-1362 (2006)

50 Duque, D.; **Vega, L.F**.
Calculation of the force between surfaces coated with grafted molecules by molecular simulation
Journal of Chemical Physics, 124, 034703 (2006)

51 Dias, A.M.A.; Carrier, H.; Daridon, J.L.; Pàmies, J.C.; **Vega, L.F.**; Coutinho, J.A.P.; Marrucho, I.M
Vapor – Liquid Equilibrium of Carbon Dioxide - Perfluoroalkane Mixtures: Experimental Data and SAFT Modeling
Industrial and Engineering Chemistry Research, 45, 2341-2350 (2006)

52 Herdes, C.; Santos, M.A.; Medina, F.; **Vega, L.F.**
Precise characterization of selected silica-based materials from Grand Canonical Monte Carlo simulations
Materials Science Forum, 514-516, 1396-1400 (2006)

53 Pratas de Melo, M.J.; Dias, A.M.A.; Blesic, M.; Rebelo, L.P.N.; **Vega, L.F**.; Coutinho, J.A.P; Marrucho, I.M.
Liquid-Liquid Equilibrium of (Perfluoroalkane + Alkane) Binary Mixtures
Fluid Phase Equilibria, 242, 210-219 (2006)

54 Herdes, C.; Lin, Z.; Valente, A.; Coutinho, J.A.P; **Vega, L.F**
Nitrogen and Water Adsorption in Aluminium Methylphosphonate alpha: A Molecular Simulation Study
Langmuir, 22, 3097-3104 (2006)

55 Pedrosa, N.; **Vega, L.F**.; Coutinho, J.A.P.; Marrucho, I.M.
Phase equilibria calculations of polyethylene solutions from SAFT-type equations of state
Macromolecules, 39, 4240-4246 (2006)

56 Llovell, F.; **Vega, L.F**.
Prediction of thermodynamic derivative properties of pure fluids through the soft-SAFT equation of state
Journal of Physical Chemistry B, 110, 11427-11437 (2006)

57 Mejía, A.; **Vega, L.F**.
Perfect Wetting along a Three-Phase Line: Theory and Molecular Dynamics Simulations
Journal of Chemical Physics, 124, 244505 (7pag) (2006)

58 Rey-Castro, C.; **Vega, L.F**.
Transport Properties of the Ionic Liquid 1-Ethyl-3-Methyl-imidazolium Chloride from Equilibrium Molecular Dynamics Simulation. The effect of Temperature.
Journal of Physical Chemistry B, 110(29),14426- 14435 (2006)

59 Llovell, F.; Peters, C.J..; **Vega, L.F**.
Second-order thermodynamic derivative properties of selected mixtures by the soft-SAFT equation of state.
Fluid Phase Equilibria, 248, 115-122 (2006)

60 J.R. Trindade, A.M.A. Dias, M. Blesic, N. Pedrosa, L.P.N. Rebelo, **L.F. Vega**, J.A.P. Coutinho and I.M. Marrucho
Liquid-liquid equilibrium of (1H,1H,7H-perfluoroheptan-1-ol + perfluoroalkane) binary mixtures
Fluid Phase Equilibria, 251, 33-40 (2007)

61 A. Olivet, D. Duque, **L.F. Vega**
Analysis of electron interactions in dielectric gases
Journal of Applied Physics, 101, 023308 (7 pag) (2007)

62 Rey-Castro, C.; Tormo, A.L.; **Vega, L.F.**
Effect of the Flexibility and the Anion in the Structural and Transport Properties of Ethyl-Methyl-Imidazolium Ionic Liquids.
Fluid Phase Equilibria, 256, 62-69 (2007)

63 A. Olivet and **L.F. Vega**
Optimized molecular force field for sulfur hexafluoride simulations
Journal of Chemical Physics, 126, 144502 (11 pag) (2007)

64 F. Llovell and **L.F. Vega**
Phase equilibria, critical behavior and derivative properties of selected n-alkane/n-alkane and n-alkane/1-alkanol mixtures by the crossover soft-SAFT equation of state
Journal of Supercritical Fluids, 41, 204-216 (2007)

65 A. M. Morales-Cas, C. Moya, B. Coto, **L. F. Vega**, G. Calleja
Adsorption of Hydrogen and Methane Mixtures on Carbon Cylindrical Cavities
Journal of Physical Chemistry C, 111, 6473-6480 (2007)

66 C. Herdes, A. Valente, Z. Lin, J. Rocha, J. A. P. Coutinho, F. Medina and **L. F. Vega**
Selective Adsorption of Volatile Organic Compounds in Micropore Aluminum Methylphosphonate-alpha. A Combined Molecular Simulation-Experimental Approach

Langmuir, 23, 7299-7305 (2007)

67 N. Pedrosa, **L. F. Vega**, J.A.P. Coutinho, I.M. Marrucho
Modeling the Phase Equilibria of Poly(ethylene glycol) Binary Mixtures with soft-SAFT EoS
Industrial and Engineering Chemistry Research, 46, 4678-4685 (2007)

68 D. Duque, B.K. Peterson, **L. F. Vega**
Interaction between coated graphite nanoparticles by molecular simulation
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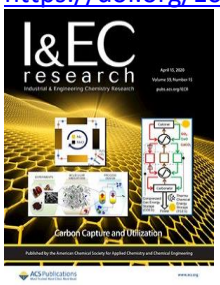

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New insights into the adsorption isotherm interpretation by a coupled molecular simulation-experimental procedure.
EDITORIAL: ISSHAC-5, ISBN 83-920343-1-7. Proceedings Fifth Internacional Symposium on Surface heterogeneity effects in adsorption and catalysis on solids, Gdansk (Poland), Pág. 63-64, (2004)

18 A.M.A. Dias, J.C. Pàmies, H. Carrier, J.L. Daridon, J.A.P. Coutinho, I.M. Marrucho,**L.F. Vega**
Critical behaviour of CO_2/perfluoroalkane mixture
EDITORIAL: ESAT2005. ISBN 83-920719-1-3. Proceedings 21st European Symposium on Applied Thermodynamics, June 1-5, 2005, Jurata, Poland. 52-55, (2005)

19 N. Pedrosa, J.C. Pàmies, **L.F. Vega**, J.A.P. Coutinho, I.M. Marrucho
The phase equilibria of ethylene glycols studied with the soft-SAFT equation of state
EDITORIAL: ESAT2005. ISBN 83-920719-1-3.Proceedings 21st European Symposium on Applied Thermodynamics, June 1-5, 2005, Jurata, Poland. 76-78, (2005)

20 A.M.A: Dias, J.C. Pàmies, **L.F. Vega**, J.A.P. Coutinho, I.M. Marrucho
Solubility of oxygen in substituted perfluorocarbons
EDITORIAL: ESAT2005
ISBN 83-920719-1-3. Proceedings 21st European Symposium on Applied Thermodynamics, June 1-5, 2005, Jurata, Poland. 130-132 (2005)

21 N. Pedrosa, **L.F. Vega**, J. A.P. Coutinho, I.M. Marrucho
Modeling of polymer phase equilibria: a soft-SAFT approach
EDITORIAL ESAT2006.
ISBN 87-91435-39-0. Proceedings 22nd European Symposium on Applied Thermodynamics, June 28-July 1st, 2006, Elsinore, Denmark, 236-239 (2006)

22 F. Llovell, J.C. Pàmies, **L.F. Vega**
Recent advances in the development and extensions of the soft-SAFT equation of state
EDITORIAL ESAT2006.
ISBN 87-91435-39-0. Proceedings 22nd European Symposium on Applied Thermodynamics, June 28-July 1st, 2006, Elsinore, Denmark, 311-314 (2006)

23 F. Llovell, A.M.A. Dias, J.A.P. Coutinho, I.M. Marrucho, **L.F. Vega**
Modeling the Perfluoralkanes Family and its Mixtures with CO2 by the soft-SAFT equation
Editorial: ESAT 2008, 443-449 (2008)

Proceedings 23h European Symposium on Applied Thermodynamics, May 29-1st June, 2008, Cannes, France

24 A. Belkadi, V. Gerbaud, **L. F. Vega**
Phase Behavior of strongly associating fluids
Editorial: ESAT 2008, 117-
Proceedings 23h European Symposium on Applied Thermodynamics, May 29-1st June, 2008, Cannes, France

25 A. Belkadi, M. K. Hadj-Kali, V. Gerbaud, X. Joulia, F. Llovell, **L. F. Vega**
Modeling the Phase Equilibria of Nitriles by the soft-SAFT Equation of State
Editorial: ESCAPE 2008 ISBN:978-0444532275
18th European Symposium on Computer Aided Process Engineering – ESCAPE 18, June 1-4, 2008, Cannes, France

26 J. S. Andreu, **L. F. Vega**
Understanding The Solubility Of Gases In Alkyl-Imidazolium Based Ionic Liquids By A Molecular Modelling Approach
Editorial: ISASF- ISBN:NN, (6 pages)
11th European International Society for Avancement of Supercritical Fluids: New Perspectives in Supercritical Fluids: Nanoscience, Materials and Processing – 11 EMSF, May, 4-7 , 2008, Barcelona, Spain

27 M.-C. Kroon, C. Herdes, **L. F. Vega**
Selective paraffin removal from olefin/paraffin mixtures by adsorption to aluminium methylphosphate-α: a molecular simulation study
Editorial: ISASF- ISBN:NN, (6 pages)
11th European International Society for Avancement of Supercritical Fluids: New Perspectives in Supercritical Fluids: Nanoscience, Materials and Processing – 11 EMSF, May, 4-7 , 2008, Barcelona, Spain

28 A. M. A. Dias, F. Llovell, J. A. P. Coutinho, I. M. Marrucho, **L. F. Vega**
Modeling the surface tension of pure alkanes and perfluoroalkanes using the soft-SAFT EoS coupled with the gradient theory extended to the critical region.
Editorial: ISASF- ISBN:NN, (6 pages)
11th European International Society for Avancement of Supercritical Fluids: New Perspectives in Supercritical Fluids: Nanoscience, Materials and Processing – 11 EMSF, May, 4-7 , 2008, Barcelona, Spain

29 F. Llovell, A. M. A. Dias, J. A. P. Coutinho, I. M. Marrucho, **L. F. Vega**
Modeing the behaviour of binary mixtures of supercritical CO2 with perfluoroalkanes by the use of crossover soft-SAFT.
Editorial: ISASF- ISBN:NN, (6 pages)
11th European International Society for Avancement of Supercritical Fluids: New Perspectives in Supercritical Fluids: Nanoscience, Materials and Processing – 11 EMSF, May, 4-7 , 2008, Barcelona, Spain

29 A. Belkadi, V. Gerbaud, F. Llovell, **L. F. Vega**
Crossover soft-SAFT modelling of the CO_2+N_2/$N2O_4$ mixture
Editorial: ISASF- ISBN:NN, (6 pages)
11th European International Society for Avancement of Supercritical Fluids: New Perspectives in Supercritical Fluids: Nanoscience, Materials and Processing – 11 EMSF, May, 4-7 , 2008, Barcelona, Spain

30 J. S. Andreu, **L.F. Vega**
Characterizing the Solubility of Gases in Ionic Liquids through a Molecular Based Equation of State
Editorial (CD):8 pages
Proceedings 17th Symposium on Thermophysical Properties, June 21 – 26, 2009, Boulder, USA.

31 A. Belkadi, M. K. Hadj-Kali, V. Gerbaud, X. Joulia, F. Llovell, **L. F. Vega**
Modeling the Phase Equilibria of Nitriles by the soft-SAFT Equation of State
Editorial: ESCAPE 2008 ISBN:978-0444532275
18th European Symposium on Computer Aided Process Engineering – ESCAPE 18, June 1-4, 2008, Cannes, France

32 R. Pacciani, A.M. López Periago, C. García-González, C. Domingo, **L. F. Vega**
A novel technique for the production of Precipitated Calcium Carbonate
Editorial (CD): 4 pages
Proceedings IV Meeting of Experts on High Pressure Supercritical Fluids Technology (FLUCOMP 2010).

33 P. López-Aranguren, C.A. García González, **L. F. Vega**, C. Domingo
Impregnation of porous substrates with alkyl- and amino-silanes using supercritical fluid technology
Editorial: Centro Stampa di Ateneo, Università degli Studi di Salerno, Fisciano (SA), Italy, September 2010, 6 pages.
Proceedings 9th International Conference on Supercritical Fluids, September 4-8, 2010, Sorrento, Italy. ISBN: 88-7897-040-9

34 C. Domingo, A.M. López Periago, R. Pacciani, A. Hidalgo-López, P. López-Aranguren, **L.F. Vega**
Impregnation Adsorbent materials for CO2 capture synthesized from scCO2: calcium carbonate and supported amines
Editorial: Centro Stampa di Ateneo, Università degli Studi di Salerno, Fisciano (SA), Italy, September 2010, 6 pages.
Proceedings 9th International Conference on Supercritical Fluids, September 4-8, 2010, Sorrento, Italy. ISBN: 88-7897-040-9

35 O. Vilaseca, F. Llovell, R. M. Marcos, **L.F. Vega**
Thermodynamic modeling of alternative refrigerants
Editorial: Centro Stampa di Ateneo, Università degli Studi di Salerno, Fisciano (SA), Italy, September 2010, 6 pages.
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36 F. Llovell, E. Valente, O. Vilaseca, R. M. Marcos, **L.F. Vega**
Thermodynamic characterization of ionic liquids and their mixtures with supercritical carbon dioxide
Editorial: Centro Stampa di Ateneo, Università degli Studi di Salerno, Fisciano (SA), Italy, September 2010, 6 pages.
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37 F. Llovell; M. B. Oliveira; O. Vilaseca; J. A. P. Coutinho; **L. F. Vega**.
Thermodynamic Modeling of the Solubility of Supercritical CO_2 and Other Gases on Ionic

Liquids with the soft-SAFT Equation of State.
Proceedings of the 10th International Symposium on Supercritical Fluids, San Francisco, USA: 13/05/2012

38 **L. F. Vega**; F. Llovell.
Simultaneous modeling of phase equilibrium and transport properties of fluids: Free-Volume Theory coupled to the soft-SAFT equation of state.
Proceedings of the IX Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE). pp. 107 - 117. 08/10/2012.

39 F. Llovell; M. B. Oliveira; O. Vilaseca; R. M. Marcos; J. A. P. Coutinho; **L. F. Vega**.
Modeling of ionic liquids: a simple and accurate approach for the prediction of thermophysical properties.
Proceedings of the IX Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE). pp. 118 - 134. 08/10/2012.

40 F Grandia, A Credoz, J Torres, R Solanas, **L Vega**
Experimental and Modelling Study of the Dry-out during CO2 SC Injection in Reservoir Rock from the Hontomn Site (Spain)
Third EAGE CO_2 Geological Storage Workshop (2012)

41 F. Llovell; O. Vilaseca; **L. F. Vega**.
Inhomogeneous density theories coupled into a molecular equation of state for the description of the fluid-fluid interface.
12th Joint European Thermodynamics Conference. 03/07/2013.

42 F. Llovell, M. B. Oliveira, J. A. P. Coutinho, R. M. Marcos, **L. F. Vega**
Ionic liquids for gas separation of greenhouse and acid gases.
4th International Congress on Green Process Engineering, 07-10/04/14.

43 M. B. Oliveira, F. Llovell, J. A. P. Coutinho, **L. F. Vega**
Modeling biodiesel production and purification: a molecular-based approach with the soft-SAFT EoS.
4th International Congress on Green Process Engineering, 07-10/04/14.

44 F. Llovell, **L.F. Vega**.
Soft-SAFT: A molecular modeling tool for the thermophysical characterization of ionic liquids.
Conference Proceedings of the 4th International Workshop on Ionic Liquids. Tarragona, Spain, 15th-16th January 2015. Pages 107-111. ISBN 84-617-3875-6.

45 Pons, M.J., **Vega, L. F.**, Olivares, L. and Guri, S.
Viabilidad de la utilización de biopolímeros como material de envase para la conservación de cerezas en atmósfera modificada.
Avances en la Postcosecha de Frutas y Hortalizas, L. Zacarías, A. R. Ballester, J. F. Marcos y M. B. Pérez-Gago (eds.) Oct 2014. Pages 229-233.

Other publications in special conference proceedings

46 Hernández-Cobos, J.; Mackie, A.D.; **Vega, L.F.**
Determination of the free energy of solvation for the dissolution of organic compounds in water at room temperature
EDITORIAL: ed. K. R: Cox, P.T. Cummings, P. R. Westmoreland. Proceedings del AIChE Topical Conference Applying Molecular Modeling and Computational Chemistry, ISBN-0-8169-9857-4. 162-163 (1998).

47 Mackie, A.D.; Hernández-Cobos, J.;**Vega, L.F.**
Vapor-liquid equilibria for an ab-initio polarizable model potential for water
EDITORIAL: ed. K. R: Cox, P.T. Cummings, P. R. Westmoreland. Proceedings del AIChE Topical Conference Applying Molecular Modeling and Computational Chemistry, ISBN-0-8169-9857-4. 195-196 (1998).

48 Blas, F.J.; **Vega, L.F.**
Prediction of critical lines and partial miscibility of binary mixtures of hydrocarbons by the soft-SAFT equation of state
EDITORIAL: ed. K. R: Cox, P.T. Cummings, P. R. Westmoreland. Proceedings del AIChE Topical Conference Applying Molecular Modeling and Computational Chemistry, ISBN-0-8169-9857-4. 303-304 (1998).

49 Blas, F.J.; Mackie, A.D.; **Vega, L.F.**
Tricritical phenomena in ternary mixtures of hydrocarbons: a SAFT approach
EDITORIAL: ed. K. R: Cox, P.T. Cummings, P. R. Westmoreland. Proceedings del AIChE Topical Conference Applying Molecular Modeling and Computational Chemistry, ISBN-0-8169-9857-4. 305-306 (1998).

50 Blas, F.J.; Gubbins, K.E.; **Vega, L.F.**
Selective adsorption fom mixtures: Grand Canonical Monte Carlo and Density Functional Theory
EDITORIAL: ed. K. R: Cox, P.T. Cummings, P. R. Westmoreland. Proceedings del AIChE Topical Conference Applying Molecular Modeling and Computational Chemistry, ISBN-0-8169-9857-4. 398-399 (1998).

51 Herdes, C.; Pàmies, J.C.; Marcos, R.M.; **Vega, L.F.**
Molecular Simulation and Theory of Associating Fluids: Phase Equilibria and Microscopic Behavior
EDITORIAL: ECCE-4th European Congress of Chemical Engineering ISBN: 84-88233-3. Proceedings of ECCE congress, Granada, vol. 5 (2003)

52 Herdes, C.; Santos, M.A.; Medina, F.; **Vega, L.F.**
Novel Approach for the Characterization and Applications of Nanomaterials Through Molecular Modeling and Experimental Adsorption Techniques
EDITORIAL: ECCE-4th European Congress of Chemical Engineering ISBN: 84-88233-3. Proceedings of ECCE congress, Granada, vol. 5 (2003)

SUPERVISED PHD THESIS

1 **Dr. Felipe J. Blas**
Title: ***Application of molecular modeling to phase equilibria calculation of multicomponent systems and charaterization of materials***.
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain
Date: 7/02/2000
Qualification: Excellent Cum Laude
PhD Extraordinary Award (given to the best PhD Thesis in the area of Science/Engineering at the University Level)

2 **Dr. Susana Figueroa-Gernstenmaier**
Title: ***Development and Application of Molecular Modelling Techniques for the Characterization of Porous Materials***
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain
Date: 13/12/2002
Qualification: Excellent Cum Laude

3 **Dr. Josep C. Pàmies**
Title: ***Bulk and Interfacial Properties of Chain Fluids: A Molecular Modeling Approach***
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain
Date: 11/12/2003
Qualification: Excellent Cum Laude
European Doctorate Recognition

4 **Dr. Fèlix Llovell**
Title: ***Accurate Description of the Critical Region by a Molecular-Based Equation of State with a Crossover Treatment***
University: Materials Science PhD Program, Universitat Autònoma de Barcelona, Spain
Date: 18/12/2006
Qualification: Excellent Cum Laude
European Doctorate recognition

5 **Dr. Carmelo Herdes**
Title: ***Rational Design of Adsorbent Materials by Molecular Simulations and Experimental Techniques***
Co-Advisor: Dr. Francisco Medina, Universitat Rovira I Virgili, Tarragona, Spain
University: Materials Science PhD Program, Universitat Autònoma de Barcelona, Spain
Date:02/2007
Qualification: Excellent Cum Laude
European Doctorare recognition

6 **Dr. Aurelio José Olivet Sopilka**
Title: ***Molecular Simulations of Sulfur Hexafluoride***
University: Materials Science PhD Program, Universitat Autònoma de Barcelona, Spain
Date: 13/05/2008
Qualification: Excellent Cum Laude

7 **Dr. Oriol Vilaseca Vidal**
Title: ***Interfacial properties of industrial relevant fluids***
Co-advisor: Dr. Fèlix Llovell, MATGAS 2000 AIE, Barcelona, Spain
University: Materials Science PhD Program, Universitat Autònoma de Barcelona, Spain

Date: 23/07/2012
Qualification: Excellent Cum Laude

8 **Dr. Santiago Builes Toro**
Title: ***Understanding the behavior of materials for capture of greenhouse gases by molecular simulations***
University: Materials Science PhD Program, Universitat Autonoma de Barcelona, Spain
Date: 14/03/2012
Qualification: Excellent Cum Laude

9 **Dr. Pedro Lopez-Aranguren**
Title: ***Functionalized adsorbent materials using supercritical CO2***
Co-advisor: Dr. Concepción Domingo, ICMAB-CSIC, Spain
University: Materials Science PhD Program, Universitat Autonoma de Barcelona, Spain
Date: 18/11/2014
Qualification: Excellent Cum Laude
International Doctorate recognition

10 **Dr. Abel Roige Godia**
Title: ***Nanoscale spatially-resolved characterization of photovoltaic devices and materials***
Co-advisor: Dr. Ramón Alcubilla, Universitat Politècnica de Calatunya, Spain
University: Materials Science and Technology PhD Program, Universitat Autònoma de Barcelona
Date: 9/02/2015
Qualification: Excellent Cum Laude
International Doctorate recognition

11 **Dr. Daniel Bahamón García**
Title: ***New generation adsorbents for CO_2 separation: from modeling to industrial applications***
University: Environmental Science and Technology Program, Universitat Autònoma de Barcelona, Spain
Date: 15/12/2015
Qualification: Excellent Cum Laude

SUPERVISED MASTER THESIS

1 **Felipe J. Blas**
Title: ***Application of molecular modeling to phase equilibria calculation of multicomponent systems and charaterization of materials***.
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain
Date: 1998
Qualification: Excellent

2 **Susana Figueroa-Gernstenmaier**
Title: ***Development and Application of Molecular Modelling Techniques for the Characterization of Porous Materials***
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain
Date: 2000
Qualification: Excellent

3 **Josep C. Pàmies**
Title: ***Bulk and Interfacial Properties of Chain Fluids: A Molecular Modeling Approach***
University: Chemical Engineering, Universitat Rovira I Virgili, Tarragona, Spain

Date: 2001
Qualification: Excellent

4 **Fèlix Llovell**
Title: ***Thermodynamic properties of Lennard-Jones chain molecules. Renormalization-group corrections to the soft-SAFT equation***
University: Graduate Program in Chemical and Process Engineering (DEA), Universitat Rovira I Virgili, Spain
Date: 5/10/2004
Qualification: Excellent

5 **Carmelo Herdes**
Title: ***Pore Size Distribution of SBA-15 and PHTS by Adsorption Data and Grand Canonical Monte Carlo Simulations***
Co-Advisor: Dr. Francisco Medina, Universitat Rovira I Virgili, Tarragona, Spain
University: Graduate Program in Chemical and Process Engineering (DEA), Universitat Rovira I Virgili, Spain
Date: 5/10/2004
Qualification: Excellent

6 **Javier Yustos Urueña**
Title: ***Molecular modeling of hydrofluorocarbons using the soft-SAFT equation of state***
University: Graduate Program in Chemical and Process Engineering (DEA), Universitat Rovira I Virgili, Spain
Date: 2005
Qualification: Excellent

7 **Aurelio José Olivet Sopilka**
Title: ***Thermodynamic and transport properties of sulphur hexafluoride predicted by molecular dynamics simulations using a simple rigid model***
University: Materials Science PhD Program (DEA), Chemistry Department, Universitat Autònoma de Barcelona, Spain
Date: 2006
Qualification: Excellent

8 **Oriol Vilaseca Vidal**
Title: ***Interfacial properties of industrial relevant fluids***
University: Materials Science Program (DEA), Universitat Autònoma de Barcelona, Spain
Date: 2009

9 **Jordi S. Andreu**
Title: ***Modeling of ionic liquids***
University: Materials Science Program (DEA), Universitat Autonoma de Barcelona, Spain
Date: 2009

10 **Pedro Lopez-Aranguren**
Title: ***First approximation for the design of adsorbent materials for CO2 capture using supercritical fl uid technology***
Co-advisor: Dr. Concepción Domingo, ICMAB-CSIC, Spain
University: Materials Science Master Program (DEA), Universitat Autonoma de Barcelona, Spain
Date: 2010

11 **Santiago Builes**
Title: ***Molecular simulations of materials for CO_2 capture***
University: Materials Science Master Program (DEA), Universitat Autonoma de Barcelona, Spain
Date: 2010

12 **Abel Roige Godia**
Title: ***New nano-materials for energy conversion***
Co-advisor: Dr. J. Oriol Ossó, MATGAS 2000 AIE

University: Nanoscience and Nanotecnology Master Program (DEA), Universitat Autònoma de Barcelona, Spain
Date: 2010

13 **Felipe Garibaldi**
Title: ***Emerging markets in the biomass field***
Co-advisor: J. Torres, MATGAS 2000 AIE, Spain
University: Master in Marketing, Universitat Autònoma de Barcelona, Spain
Date: 2013

14 **Andreu Oliver**
Title: ***Strategic marketing analysis of water treatment for recalcitrant contaminants in water***
Co-advisor: Dr. S. Guri, MATGAS 2000 AIE, Spain
University: Master in Marketing, Universitat Autònoma de Barcelona, Spain
Date: 2013

15 **Yiyang Li**
Title: ***Marketing analysis and needs for new energy solutions in China***
Co-advisor: Dr. U. Renh, Air Products and Chemical, USA
University: Master in Marketing, Universitat Autònoma de Barcelona, Spain
Date: June 2013

16 **Wanli Ma**
Title: Agrifood Opportunities in China
Co-advisor: Dr. S. Guri, MATGAS 2000 AIE
University: Master in Marketing, Universitat Autònoma de Barcelona, Spain
Date: June 2014

17 **Laura Gelabert**
Title: ***Environmental Benefits of alternative fuel for public transportation- a comparative LCA of hydrogen and electricity***
University: Master in Environmental Science and Technology, Universitat Autònoma de Barcelona, Spain
Date: 30/06/2014

18 **Xavier Llavina**
Title: ***Life Cycle Analysis of delivering alternative fuels: the hydrogen and electricity cases in Great Britain***
University: Master in Environmental Science and Technology, Universitat Autònoma de Barcelona, Spain
Date: 25/06/2014

19 **Anna Mateo**
Title: ***Molecular Dynamics of gases diffusion in zeolites for separation purposes***
University: Master in Industrial Chemistry and Introduction to Chemical Research, Universitat Autònoma de Barcelona, Spain
Date: 10/09/2015

20 **Eduard Cané**
Title: ***Application of molecular modeling tools to Oil & Gas transport properties***
Co-advisor: Dr. Felix Llovell, Institut Quimic de Sarriá, URL, Spain.
University: Master in Chemical Engineering, Institut Quimic de Sarrià. Universitat Ramon Llull, Spain. In collaboration with the Petroleum Institute of Abu Dhabi., UAE.
Date: June 2017

21 **Mostafa Elabyouki**
Title: ***Molecular simulations of nature-inspired hierarchical porous materials for energy storage***
Co-advisor: Dr. Maryam Khaleel, Khalifa University, UAE.
University: Master of Science in Chemical Engineering. Khalifa University of Science and Technology, Abu Dhabi, UAE.

Date: May 2018.

22 **Yuting Li**
Title: ***Phase and interfacial properties of low global warming potential refrigerants by molecular dynamics simulations***
Co-advisor: Dr. Wael A. Fouad, Khalifa University, Abu Dhabi, UAE.
University: Master of Science in Chemical Engineering. Khalifa University of Science and Technology, Abu Dhabi, UAE.
Date: May 2018.

23 **Enrique Pelegrin**
Title: ***An in-depth study of a simulated adsorption process using different adsorbents***
Co-advisor: Dr. Felix Llovell, Universitat Ramon Llull (URL), Barcelona, Spain.
University: Master in Chemical Engineering. Institut Quimic de Sarria, URL, Barcelona, Spain
Date: July 2018

24 **Jose Justino**
Title: ***Thermodynamic and Interfacial properties of hydrogenated and fluorinated alcohols mixtures: A molecular modelling approach***
Co-advisor: Dr. Eduardo J.M. Filipe, Instituto Superior Tecnico, Universidade de Lisboa, Portugal.
University: Master of Science in Chemical Engineering. IST, Lisbon, Portugal
Date: November 2018

25 **Hashir I. Nasim**
Title: ***Understanding the corrosion inhibitor film on metal surfaces by experiments and molecular simulations.***
Co-advisor: Dr. Ricardo Nogueira, Khalifa University, Abu Dhabi, UAE.
University: Master of Science in Chemical Engineering, Khalifa University of Science and Technology, Abu Dhabi, UAE.
Date: December 2018.

26 **Cristina Martinez**
Title**: *Understanding the behaviour of aqueous amines for CO2 capture in the presence of degraded amine products by experimental analysis and computational methods.***
Co-advisors: Dr. Fèlix Llovell, IQS, Barcelona, Spain. Dr. Kyra Lauren Sedransk Campbell, Imperial College, London, UK.
University: Master of Science in Chemical Engineering, IQS, Universitat Ramon Llull, Barcelona, Spain.
Date: July 2019.

27 **João Eduardo Duarte**
Title: ***Modelling Phase Equilibria, Transport and Interfacial Properties of Mixtures of Fluorinated and Hydrogenated Alcohols.***
Co-advisor: Dr. Eduardo Filipe, Chemical and Biological Engineering, Instituto Superior Técnico, Universidade Técnica de Lisboa, Portugal.
University: Master of Science in Chemical Engineering. IST, Lisbon, Portugal.
Date: November 2019.

28 **Seba Alareeqi**
Title: New Insights into the Effect of CO2 and NaCl Aqueous Solution on the Adsorption Behavior of Corrosion Inhibitors through Molecular Simulations.
Co-advisor: Dr. Ricardo Nogueira, Khalifa University, Abu Dhabi, UAE
University: Master of Science in Chemical Engineering, Khalifa University of Science and Technology, Abu Dhabi, UAE
Date: November 2019.

PARTICIPATION IN CONFERENCES

A total of **more than 600 presentations in international conferences** and workshops.
~70 invited/plenary and 150+ additional presentations since 2011.
Chairperson of the International SAFT Series of conferences (2003, 2010, 2015,2017,-).
Organizer (chairperson) of 8 scientific conferences. Chair of several sessions in international conferences.
Member of the **international scientific committee** of five conferences related to thermodynamics and clean energy.

PLENARY AND INVITED PRESENTATIONS (2011-2020)

2011

1. **Renormalization Group Theory Combined with Molecular-Based Equations of State: An Engineering Tool with the Right Physics Inside.** L.F. Vega. Invited seminar "In Honor of Jan Sengers' 80th Birthday". AIChE Annual Meeting Minneapolis Convention Center, Minneapolis, October 18th, 2011.
2. **Challenges and opportunities for a sustainable energy scenario: the key role of research and development in chemistry**, Invited Lecture. L.F. Vega. Workshop on Research and Innovation in Chemistry and Energy, Campus of International Excellence Southern Catalonia (CEICS). Tarragona, Spain. November 9-11th, 2011.
3. **The CO_2 as a resource, from capture to industrial applications**. S. Guri and L.F. Vega. Seminar in the framework of the course "Capture, transport, storage and CO_2 utilization". Madrid, Spain. 28th November to 1st December, 2011.

2012

4. **CO_2, a problem or an opportunity?** L.F.Vega, invited seminar at the Universitat de Girona, as part of the seminar series Interdisciplinarium. Universitat de Girona, Spain. February 22nd, 2012.
5. **MATGAS & Food Research in its laboratories**. Lourdes F. Vega, presentation at the 7a Food Technical Day about gases utilization in the food industry: freezing, chilling and food packaging: Organized by Carburos Metálicos- Air Products Group- MATGAS-UPV. Parque Científico de la Universidad Politécnica de Valencia, Valencia, Spain. February 8th, 2012.
6. **Effective Energy.** L.F.Vega, invited lecturer at EULASUR Workshop. Copenhagen Business School, Copenhagen, Denmark, March 14th, 2012.
7. **Nano-Technologies, Jobs and Start-Up** - L.F.Vega. Roundtable at EULASUR Workshop - Copenhagen Business School. Copenhagen, Denmark. March 14th, 2012.
8. **Scientific investigation in the private business world.** L.F.Vega, invited lecturer at the course U2B: From University to Business, by the UAB. Universitat Autònoma de Barcelona. March 31st, 2012.
9. **Using molecular-based equations of state for industrial relevant problems: the importance of having the right physics.** L.F. Vega, invited seminar at the IFP Energies Nouvelles of Paris. IFPEN, Paris-Cedex, France, April 11th, 2012.
10. **Food: Sustainability, innovation and future,** L.F. Vega, Conference given as part of the conference-dinner offered by Carburos Metálicos, Air Products Group, in the framework of the Exhibition Barcelona Tecnoalimentaria. Barcelona, Spain, May 15th, 2012.
11. **Materials for carbon capture and utilization in the context of sustainable energy development.** L.F. Vega, Invited Seminar given at the University of Granada, Applied Physics Department, Granada, Spain. May 24th, 2012.
12. **Research, Environment and Sustainibility.** L. F. Vega, invited speaker at the thematic roundtable "Innovation and Sustainability" at the ANQUE's International Congress of Chemical Engineering (ANQUE-ICCE 2012), Seville, Spain. June 24-27th, 2012.

13. **CO_2 Capture, Sequestration and Utilisation: A Key Player in the Sustainable Energy Scenario**, L.F. Vega invited speaker at the thematic Roundtable "Carbon, Capture and Storages Technologies or Sequestration" at the BIT's 1st Annual International Symposium of Clean Coal Technology - CCT-2012, Taiyuan, China. September 24-26th , 2012.
14. **Extending the Capabilities of Molecular-Based Equations of State: Simultaneous Calculations of Phase Equilibria and Transport Properties.** L.F. Vega, F. Llovell, R.M. Marcos, Invited talk at 2012 AICHE Annual Meeting, Pittsburgh, USA, October 28th – November 2nd, 2012.

2013

15. **Understanding supercritical CO_2: from fundamental to industrial applications**. L. F. Vega, S. Builes, F: Llovell, J. Torres, R. Solanas, P. López-Aranguren, C. Domingo, O. Vilaseca, R.M. Marcos. Invited speaker at the 3rd International Solvothermal and Hydrothermal Association Conference, Austin, Texas (USA), January 13-17th, 2013.
16. **Supercritical CO_2 in chemical processes: functionalisation of porous silica with alkoxysilanes.** P. López-Aranguren, L.F. Vega, C. Domingo, Faculty of Science, Monash University, Melbourne, Australia. February 21st, 2013.
17. **Innovation in Chemical Processes to make them more sustainable.** L.F. Vega. Aragon Institute for Engineering Research (I3A). University of Zaragoza. Spain. March 22nd 2013.
18. **Capture and Utilization of Carbon Dioxide (CO_2): Turning challenges into business opportunities.** L.F. Vega. Energy and Environmental Seminar. Fundacion Gas Natural Fenosa, Madrid, Spain. June 5th, 2013**.**
19. **Inhomogeneous Density Theories Coupled into a Molecular Equation of State for the Description of the Fluid-Fluid Interface.** F. Llovell, O. Vilaseca, L. F. Vega. 12th Joint European Thermodynamics Conference. Brescia, Italy, July 3rd, 2013.
20. **Modeling thermophysical properties of industrial relevant systems: molecular simulations and thermodynamic perturbation theory.** A. García-Sanchez, F. Llovell, S. Builes, L.F. Vega. XXIX Annual Meeting of the Reference Network of R+I on Theoretical and Computational Chemistry. Bellaterra. Barcelona, Spain. July 10th -11th, 2013.
21. **"Un viatge virtual al món de les moléculas" (A virtual tour to the molecular world)**. X. Giménez, J-D Marechal, L.F. Vega. XXIX Annual Meeting of the Reference Network of R+I on Theoretical and Computational Chemistry. Bellaterra. Barcelona, Spain. July 10th -11th, 2013.
22. **CO_2 Utilization.** J. Torres, L.F. Vega. Course in CCS Technologies: Capture, Transport and geological storage of CO2. Ponferrada, Leon, Spain. July 19th, 2013.
23. **Optimized adsorbent materials for CO_2 capture and separation: advanced experimental techniques and molecular simulations**. Invited talk. L.F. Vega. International Symposium on CO_2 capture: microscopic studies and applications". University Paris-Est Marne la- Vallee, France. 18-20th September 2013.
24. **Sustainable water management and open innovation.** Invited talk. S. Guri, L.F. Vega. ChemH_2O2013- Madrid, Spain, 1-2 October 2013.
25. **Latest development of thermodynamic models: the case of SAFT and its refined version for industrial applications**. L. F. Vega. TOTAL 21th Corporate Process Engineering Meeting (PEM). Le Touquet, France. October, 15 -16th, 2013.
26. **Challenges and opportunities using supercritical CO_2: from fundamentals to industrial applications**. Plenary talk. L. F. Vega. Workshop on Supercritical Fluids and Energy. Campinas, Brazil. December, 8-11th, 2013.

2014

27. **CO_2 Capture, Sequestration and Utilization: A Key Player in the Sustainable Energy Scenario.** L.F. Vega. NETWORKING NATIONS II: Scientific Opportunities in UK and Spain. Madrid, Spain. January 21st 2014.

28. **Supercritical fluids: a field in physics with multiple applications.** Vega, L.F. Invited lecturer at the Conference Cycle: "Science and Technology in the XXI Century", organized by the Academy of Science, Mathematics, Physics, Chemistry and Natural Sceinces of Granada, March 26th, 2014, Granada, Spain.

29. **SAFT and Industry: challenges and opportunities.** Vega, L.F. "Plenary talk at the SAFT Meeting 2014: From SAFT to Experiments and Back. Tróia, Portugal, April 22nd -24th, 2014.

30. **Uses of CO_2**. L. F. Vega, Round table session at FLUCOMP 2014. Barcelona, Spain, June 10-13th 2014.

31. **Research and Business: Collaborate to better compete**. L.F.Vega. Research and Innovation Roadmap Presentation at the Parliament of Catalonia. Barcelona, Spain. June 20th 2014.

32. **Thermodynamic modeling: challenges and opportunities in moving it from academia to industry and backwards**. Vega, L.F. Plenary talk at the TERMO 2014 – GET, XIV Meeting of the specialized group in Thermodynamics, Spanish Royal Society of Physics and Chemistry, Baiona, Vigo, Spain. September 14-16th 2014.

2015

33. **Soft-SAFT: a molecular modeling tool for the thermophysical characterization of ionic liquids**. Invited talk in honor of Prof. H. Segura. L.F. Vega, F. Llovell. 4th International Workshop on Ionic Liquids. Tarragona, Spain, 15th-16th January 2015.

34. **Research and technology transfer in a public institution or in a company?** Invited lecture. Postgraduate course From University to Business (U2B). Universitat Autònoma de Barcelona, Spain. February 14th, 2015.

35. **Rountable entitled "Uses of CO_2: Food and Water".** L.F.Vega. Mediator at the Worskhop Aportando Valor al CO2, organized by Worskhop Aportando Valor al CO_2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

36. **Innovation in CO_2 Capture Process – Sustainability and CO_2?.** Invited Speaker. L.F. Vega. Workshop "Spanish Innovation for Renewable Energies: the future of renewable energies as solution for sustainability". Deusto Business School, Global Center for Sustainable Business. Madrid, March 3rd 2015.

37. **Applications of Carbon Dioxide in Bioenergy production**. Invited lecturer. L.F. Vega. "1st Bioenergy Conference". Universidad de Salamanca, Bioenergy Association of Salamanca (ABSal). Salamanca, March 4th 2015.

38. **CO_2 uses in the context of sustainable development: challenges and opportunities.** Plenary talk. L.F. Vega. IMM2015. VII International Conference on Mechanical Engineering, Cartagena de Indias, Colombia, 28-30 April, 2015.

39. **SAFT for associating fluids: successes and remaining challenges.** Plenary talk. L. F. Vega. A 25th anniversary celebration of the Statistical Associating Fluid Theory (SAFT). SAFT2015. Houston, USA, May 17-19, 2015.

40. **Latest development of thermodynamic models for oil and gas industries: the SAFT model**. Invited Seminar. Corporate and Innovation Technology Department, Halliburton, Houston, USA. May 22nd, 2015

41. **Assessing gas separation of greenhouse and acid gases with ionic liquids**. Plenary talk. EQUIFASE 2015, Alicante, Spain. June 28th, July 1st, 2015.

42. **Greenhouse and Acid Gases with Ionic Liquids - Predictive Model**. Invited talk. Iberoamerican Meeting on Ionic Liquids (IMIL2015), Madrid, Spain. July 2-3, 2015.

43. **Challenges and opportunities using Statistical Mechanics modeling in the industrial world.** Keynote Lecture. L.F. Vega, V Summer School on Statistical Physics of Complex and Small Systems, CRM Colloquium. Barcelona, July 7th, 2015.

44. **Materials for CO_2 capture and other industrial applications: synergies between molecular simulations and experiments.** Plenary talk. L.F. Vega. 6th Iberian Meeting on Colloids and Interfaces Gimerães, Portugal. July 8-10, 2015.

45. **Synergies between simulations and experiments for developing sustainable processes: the case of CO_2 capture and utilization.** Invited seminar. L.F. Vega. Masdar Institute, Abu Dhabi, UAE, 8th December 2015.

46. **CO_2 capture and utilization: where are we today and where are we going?.** Invited talk. L.F. Vega. CO2 capture and utilization, Workshop organized by Xermae and XRQTC, Barcelona, Spain, 17th December 2015.

2016

47. **Searching for sustainable energy. Searching for solutions.** Invited seminar. L.F. Vega. Masdar Institute, Abu Dhabi, UAE, 12th April 2016.

48. **SAFT as a tool for phase equilibria calculations and process design: advances and remaining challenges,** Plenary talk, 14th PPEPPD, International Conference on Properties and Phase Equilibria for Product and Process Design, Porto, 22-26 May 2016.

49. **How can molecular modeling tools help in the sustainable energy scenario?**, Invited talk, Joint EMLG/JMLG Annual Meeting 2016, Recent Progresses on the Experimental & Theoretical-Computational Techniques for the Study of Liquids and Supercritical Fluids. From Simple to Complex Systems, Platanias - Chania, Crete, Greece, 11-16 September 2016.

50. **Recent advances and remaining challenges in materials and processes for CO2 Capture and Utilization**, Invited talk, EU-GCC Clean Energy Network II. Joint Working Groups meetings on Carbon Capture and Storage and on Clean Gas, The Petroleum Institute, Abu Dhabi, UAE, 7 November 2016.

51. **Modeling of alternative solvents for sustainable processes: supercritical CO2 and ionic liquids**. Invited talk in honor of Prof. C.J. Peters. L.F. Vega, F. Llovell. AIChE Annual Meeting. AICHE2016. San Francisco, California, USA, November 13-18, 2016.

52. **New procedure for enhancing SAFT molecular parameters transferability: the role of derivative properties**. Invited talk in honor of Prof. T. de Loos. L.F. Vega, F. Llovell, M.B. Oliveira, J.A.P. Coutinho. AIChE Annual Meeting. AICHE2016. San Francisco, California, USA, November 13-18, 2016.

53. **Molecular modeling and advanced experimental techniques – from fundamentals to product and process design and implementation**, Invited talk, PIRC R&D Conference and Exhibition, The Petroleum Institute, Abu Dhabi, UAE, November 21st, 2016.

2017

54. **Molecular Modeling as a helpful tool in Product and Process Design**. Invited seminar. TAKREER Research Center. Knowledge Sharing Program, Abu Dhabi, UAE, January 11th, 2017.

55. **Molecular modeling tools in the sustainable energy scenario**. Guest invited speaker. 4th CNIE Advisory Board Meeting and Workshop, University College of London, UK, January 30-31, 2017.

56. **Application of molecular modeling techniques to oil and gas**. Invited presentation. Workshop on Asphaltenes. The Petroleum Institute, Abu Dhabi, February 12th, 2017

57. **Chemical engineers of the 21st century- the dilemma of what to do and the needed flexibility to change and to adapt**. Invite talk in the conference "Catalyse your future- JEQ", Instituto Superior Tecnico de Lisboa, Portugal, March 14th, 2017.

58. **Soft-SAFT Equation of State Applied to Oil & Gas Mixtures**. Plenary talk. EQUIPETRO (1st Meeting on Phase Equilibria in Oil & Gas). Salvador Bahia, Brazil, March 22-24, 2017.

59. **CO_2 capture, transport and utilization – accomplishments and challenges**, Invited talk. Workshop on"CO2 CAPTURE, UTILIZATION AND STORAGE – WHERE ARE WE?". The Petroleum Institute, Abu Dhabi, April 19th, 2017.

60. **Dr. Jekyll and Mr. Hyde: the two faces of CO_2.** Invited speaker. Ateneo of Zaragoza 150th special contribution. Universidad de Zaragoza, Spain, April 26th, 2017.

61. **Innovación abierta /Open Innovation – how to use the needs to generate good innovation ideas.** Plenary talk – inauguration of the Master in R&D Management and in Chemical Engineering at the Universidad de Zaragona, Spain, April 27th, 2017.

62. **Towards a predictive tool for PVT characterization: what can we learn by using SAFT versus classical approaches**. Invited talk in the Seminar Series "Cup of Tea with PVT", ADCO company, Abu Dhabi, UAE, May 23rd, 2017.

63. **Climate Change, greenhouse gases and what to do with them.** Reception invited presentation at the Official Ceremony of Entrance at the Academy of Mathematics, Physical- Chemistry and Natural Sciences of Granada, as an Elected Corresponding member, October 6th, 2017.

64. **Understanding the Removal of Pharmaceuticals from Water Effluents By Adsorption in Activated Carbons – a Molecular Simulation Approach**. Invited talk in the Special Session in Honor of Keith Gubbins' 80th birthday, AIChE Annual Meeting, Minneapolis, MI, USA, October 29 – November 3, 2017

2018

65. **Research & Development and Market needs: effective management of R&D in companies.** Invited plenary lecture at the Graduate Course in R&D Management and the Master in Chemical Engineering at the Universidad de Zaragona, Spain, April 18th, 2018.

66. **Searching for sustainable energy: from molecules to processes.** Invited seminar at Chemistry Department, New York University Abu Dhabi, UAE, May 1st, 2018.

67. **Kigali agreement: Action on the emissions of fluorinated greenhouse gases outside the SUDOE area**. Invited keynote lecture at the KET4F-Gas Launch Event, Caparica, Portugal, June 11th, 2018.

68. **The role of thermophysical properties knowledge in mitigating climate change.** Plenary talk at the Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29th, 2018.

69. **How can the theory of complex fluids help in mitigating climate change?** Plenary talk at the Foundations of Molecular Modeling and Simulations (FOMMS) Conference, Delavan, Wisconsin, WI, USA, July 15-20th, 2018.

2019

70. **From Molecules to Processes: Designing Effective Clean Energy Processes from Understanding Molecular Interactions.** Invited seminar at the IQS School of Engineering, Universitat Ramon Llull, Barcelona, Spain, January 16th, 2019

71. **Research and Innovation Center on CO_2 and H_2**. Invited presentation at the UAE-Japan Industry-Academia Collaboration Steering Board Meeting, Khalifa University, Abu Dhabi, UAE, February 18th, 2019.

72. **Research & Development in industry and the importance of knowing market needs.** Invited plenary lecture at the Graduate Course in R&D Management and the Master in Chemical Engineering at the Universidad de Zaragona, Spain, March 28th, 2019.

73. **Climate Change: what to do? Are we running out of time?** Invited lecture at the Academy of Mathematics, Physical Chemistry and Natural Sciences of Granada, Spain, April 1th, 2019.

74. **Development of next generation refrigerants: challenges and opportunities**. Keynote lecture at the *Workshop Mitigation of the environmental impact of fluorinated gases*, 3rd International Conference on Chemical Engineering (ANQUE-ICCE 3), Santander, Spain, June 19-21st, 2019. ISBN: 978-84-09-12430-5, page 381.

75. **Molecular simulations as a tool to design carbon-based materials for sustainable processes**. Keynote lecture at the *Iberoamerican Symposium on Materials by Design: From the Nano-scale to Engineered Materials* (ANQUE-ICCE3-CIBIQ), Santander, Spain, June 19-21st, 2019. ISBN: 978-84-09-12431-2, page 236.

76. **Invited speaker at the roundtable "Innovation and Future"**, 3rd International Conference on Chemical Engineering (ANQUE-ICCE 3), Santander, Spain, June 19-21st, 2019.

77. **The role of H_2– a UAE and global outlook. Why a center on CO_2 and H_2 at KU**, Keynote lecture/Inaugural lecture at the First UAE-Japan Hydrogen Workshop, Abu Dhabi, 15th December 2019.

2020

78. **A molecular perspective on the search for efficient processes for CO_2 capture and separation - the role of modeling**, Keynote lecture at the 1st Iberian Conference on Supercritical Fluids (former FLUCOMP) conference organized in parallel with the "International Conference on Aerogels for Biomedical and Environmental Applications", a conference organized by the AERoGELS COST Action, Santiago de Compostela, Spain, 18-19 February 2020.
79. **Women in Power,** Invited panelist at the event in the framework of the Middle East Electricity (MEE) 2020, under the patronage of H.H. Sheikh Maktoum bin Mohammed bin Rashid Al Maktoum, Dubai Deputy Ruler, Dubai World Trade Centre, Dubai, UAE, 3-5 March 2020
80. **Understanding the performance of hierarchical carbon electrodes for lithium-air batteries by molecular dynamics simulations, (co-authors:** Lourdes F. Vega, M. Elabyouki, D. Bahamon, M. Khaleel) Invited lecture at the International Conference on Nanomaterials (e-conference), Sofia, Bulgary (online), 19-20 May 2020.
81. **Open Innovation: from research to market**. Invited lecture in the graduate course "Introduction to Research in Chemical Engineering", Expertia program. Master in Chemical Engineering and Environmental Technologies (remote connection). Zaragoza. Spain, 22 May 2020.

ADDITIONAL CONTRIBUTIONS: ORAL AND POSTER PRESENTATIONS (2011-2020)

2011

1. **Sorption of tryalkoxysilane in low-cost porous silicates using a supercritical CO_2 method.** P. López-Aranguren, J. Saurina, L.F. Vega, C. Domingo. 13th European Meeting on Supercritical Fluids Congress. The Hague (The Netherlands). October 9 – 12th, 2011.
2. **Materials for CO2 capture prepared using supercritical CO_2.** A. M. López-Periago, P. López-Aranguren, R. Pacciani, L.F. Vega, C. Domingo. 13th European Meeting on Supercritical Fluids Congress. The Hague (The Netherlands). October 9 – 12th, 2011.
3. **Molecular Simulations of Tryalkoxysilane Functionalized Porous Silicates.** S. Builes, P. Lopez-Aranguren, R. Pacciani, C. Domingo and L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.
4. **Sustainable Energy Bio-Sources, an Environmental Performance Approach: Life Cycle Assessment From WWTP to BESs.** G. Blejman, R. Pacciani, S. Guri and L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.
5. **Critical, Interfacial and Derivative Properties of Ionic Liquids by a Molecular-Based Equation of State.** O. Vilaseca, F. Llovell, L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16-21st, 2011.
6. **Adsorption of CO_2 Using Amine Functionalized Silica.** S. Builes, L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011. 7. Renormalization Group Theory Combined with Molecular-Based Equations of State: An Engineering Tool with the Right Physics Inside. L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.
7. **Distributed Model Predictive Control for UK CO_2 Capture, Transport and Storage Network.** A. Arce, N. Mac Dowell, Alejandro J. del Real, N. Shah, L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.
8. **The Application of a Model Predictive Control-Based Approach to CO_2 Capture Processes: Towards Operational Cost Minimisation.** A. Arce, N. Mac Dowell, N. Shah, L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.
9. **Towards Green Bioprocessing: Ionic Liquids for Biomass Deconstruction.** N. Mac Dowell, J. P. Hallett, O. Vilaseca, F. Llovell, L. F. Vega. AIChE Annual Meeting 2011, Minneapolis (USA). October 16 – 21st, 2011.

10. **Molecular modeling of aqueous systems using the soft-SAFT EoS.** F. Llovell, O. Vilaseca, L. F. Vega. SAFT 2011. Pau (France). October 24 – 25th, 2011.
11. **Modeling Ionic Liquids with soft-SAFT: an exploration of critical, interfacial and derivative properties**. Oriol Vilaseca, Felix Llovell, Lourdes F. Vega. SAFT 2011. Pau (France). October 24 – 25th, 2011.
12. **Synthesis of Biodiesel Using Supercritical CO_2 as Solvent: Lipids Recovery From Frying Oil.** J. Torres, R. Solanas and L.F. Vega (poster presentation). 12th Mediterranean Congress of Chemical Engineering. Barcelona (Spain). November 15-18th, 2011.
13. **Thermodynamic modeling of ionic liquids for separation and extraction processes.** F. Llovell, L.F. Vega. 12th Mediterranean Congress of Chemical Engineering. Barcelona (Spain). November 15 – 18th, 2011.
14. **Characterization of thermophysical properties of blends of refrigerants using a molecular-based equation of state.** F. Llovell, L.F. Vega (poster presentation). 12th Mediterranean Congress of Chemical Engineering. Barcelona (Spain). November 15 – 18th, 2011.
15. **Modified Atmosphere Packaging as an alternative for controlling pests in food products.** S. Guri, M.J. Pons, L. F. Vega and J. Riudavets. Eurocereal Conference. Campden BRI (United Kingdom). December 6 – 7th, 2011.

2012

16. **Thermally-induced morphological changes and nanoscale electrical properties in semiconducting organic thin films.** A. Roigé, J. O. Ossó, M. Campoy-Quiles, and L. F. Vega (poster presentation), EULASUR Workshop, Business School of Copenhagen, Copenhagen (Denmark). March 14-16th , 2012.
17. **From waste to energy. Converting wet biomass into energy by oxygen enriched air.** J. Torres, L. F. Vega, M. Hevey, A. Slavejkov, M. D'Agostini, S. Poussou, X. He, J. Godoy and C. Esquius. Air Products 2012 Growth conference. Growth successes poster session. Allentown (USA). April 18th, 2012.
18. **Towards a fundamental understanding of the kinetics of carbon formation.** R. Pacciani, J. Torres, R. Ramirez, D. Garg, L. Vega, E. Klingenberg. 2012 Growth conference. Growth successes poster session. Allentown (USA). April 18th, 2012.
19. **Thermodynamic Modeling of the Solubility of Supercritical CO_2 and Other Gases on Ionic Liquids with the soft-SAFT Equation of State.** F. Llovell, M. B. Oliveria, O. Vilaseca, J.A.P. Coutinho, L. F. Vega. Presented at the 10th International Symposium on Supercritical Fluids, San Francisco, CA, (USA) May 13-16th 2012.
20. **New approach to control fuel cell stacks under variable loads.** A. Arce, L.F. Vega, C. Bordons (oral presentation). 3rd LOLIPEM International Workshop: Fuel Cell Stacks. Altafulla (Spain). May 29-30th, 2012.
21. **The AFM as a characterization tool for polymeric electrolyte membranes.** I. Torres, A. Arce, J. O. Ossó, L. F. Vega, M. L. DiVona (oral presentation). 3rd LOLIPEM International Workshop: Fuel Cell Stacks. Altafulla (Spain). May, 29-30th 2012.
22. **Modeling the [Tf2N] pyridinium Ils family and their mixtures with the soft-SAFT EoS.** M.B. Oliveira, F. Llovell, J.A.P. Coutinho and L. F. Vega. Presented at the ANQUE's International Congress of Chemical Engineering (ANQUE-ICCE 2012), Seville (Spain). June 24-27th 2012.
23. **Surface Tension of Binary Mixtures of Ionic Liquids with the Common Bistriflamide Anion: Experimental Measurements and Modeling with the soft-SAFT EoS.** M. Domínguez-Pérez, M.B. Oliveira, M. G. Freire, F. Llovell, O. Cabeza, J. A. Lopes da Silva, L. F. Vega, J. A. P. Coutinho. Presented at the ANQUE's International Congress of Chemical Engineering (ANQUE-ICCE 2012), Seville (Spain). June 24-27th 2012.
24. **Transferable Molecular Models for the Calculation of Thermodynamic and Transport Properties of Common Hydrofluorocarbon Refrigerants and New Alternative Blends with the soft-SAFT Equation.** O. Vilaseca, F. Llovell, R.M. Marcos, L.F. Vega. Presented at the 18th Symposium on Thermophysical Properties, University of Colorado Campus, Boulder, Colorado (USA). June 24-29th 2012.

25. **Describing the Absorption of Weak Electrolytes and Acid Gases with Ionic Liquids Using the soft-SAFT Approach.** F. Llovell, R.M. Marcos, N. MacDowell, L. F. Vega. Presented at the 18th Symposium on Thermophysical Properties, University of Colorado Campus, Boulder, Colorado (USA). June 24-29th 2012.
26. **Accurate Thermophysical Characterization of Aqueous Systems with the soft-SAFT Equation of State.** F. Llovell, O. Vilaseca, R.M. Marcos, L.F. Vega. Presented at the 18th Symposium on Thermophysical Properties, University of Colorado Campus, Boulder, Colorado (USA). June 24-29th 2012.
27. **Application of a Renormalization-Group Treatment to the Statistical Associating Fluid Theory for Potentials of Variable Range (SAFT-VR).** E. Forte, F. Llovell, L. F. Vega, J. P. Martin Trusler and A. Galindo. Presented at the 18th Symposium on Thermophysical Properties, University of Colorado Campus, Boulder, Colorado (USA). June 24-29th 2012.
28. **Research and Development in water treatment at MATGAS and Carburos Metálicos-Air Products Group: towards sustainable water solutions**. W. Chang, L. F. Vega, S. Guri. (poster presentation), Ecotechnologies for Wastewater Treatment Congress, University of Santiago de Compostela (Spain). June 25 – 27th 2012.
29. **Characterization of the behavior of geological materials in contact with supercritical CO_2.** R. Solanas, A. Credoz, J.Torres, F. Grandia, J. Bruno and L.F. Vega. FLUCOMP 2012 VII Meeting of experts in compressed fluids tecnologies, Madrid (Spain). June 28-29th 2012.
30. **Using molecular simulation to tailor optimal adsorbent and catalytic materials for specific applications**. A.Garcia-Sanchez. Global Technology Centers Seminar Series. Allentown (USA,). July 11th 2012.
31. **Experimental and theoretical comparison of an equation of state for pore-confined fluids**. Domingo, C., Lopez-Aranguren, P., L.F. Vega. Presented at the 4th Congress of the European Association for Chemical and Molecular Sciences, Prague, (Czech Republic. August 26 – 30th , 2012
32. **Kelvin probe force microscopy for photovoltaic materials characterization**. A. Roigé, J. O. Ossó, C. Voz, R. Alcubilla, L. F. Vega. CFN Summer School 2012 on Nano-Energy. Bad-Herrenalb (Germany), September 14-17, 2012.
33. **New forward osmosis membranes with internal morphology design**. K. Briceño, I. Fonseca, L. Vega, S. Guri, R. Garcia-Valls (poster presentation). Euromembrane 2012 Conference. London (UK). September 23-27th 2012.
34. **Sorption of tryalkoxysilane in low-cost porous silicates using a supercritical CO_2 method.** P. López-Aranguren, J. Saurina, L.F. Vega,C. Domingo. 13th European Meeting on Supercritical Fluids Congress. The Hague (The Netherlands). October 9 – 12th, 2012.
35. **Surface Tension of Binary Mixtures of 1-Alkyl-3-methylimidazolium Bis(trifluoromethylsulfonyl)imide Ionic Liquids: Experimental Measurements and soft-SAFT Modeling.** M.B. Oliveira, M. Domínguez- Pérez, M.G. Freire, F. Llovell, O. Cabeza, J.A. Lopes-da-Silva, L.F. Vega and J.A. P. Coutinho. Awarded as "the best poster" in EQUIFASE 2012 Conference. Puerto Varas, Chile, October 8th – 12th, 2012.
36. **Solubility of CH_4, N_2, CO_2 and N_2O in 1-ethyl-3-methylimidazolium methyl-phosphonate –soft-SAFT Modelling and Selectivities.** L. Pereira; P. J. Carvalho; M. B. Oliveira; F. Llovell; L. F. Vega; J. A. P. Coutinho. IX Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE), Puerto Varas, Chile, October 8th – 12th, 2012.
37. **Modeling of ionic liquids: a simple and accurate approach for the prediction of thermophysical properties.** F. Llovell, M. Belo, O. Vilaseca, R.M. Marcos, J.A.P. Coutinho, L.F. Vega. IX Iberoamerican Conference Phase Equilibria and Fluid Properties for Process Design – Equifase 2012. Puerto Varas, Chile, October 8th – 12th, 2012.
38. **Application of a density functional theory to the prediction of interfacial properties of complex mixtures.** F. Llovell; N. Mac Dowell; F. J. Blas; A. Galindo; G. Jackson. IX Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE), Puerto Varas, Chile, October 8th – 12th, 2012.
39. **Simultaneous modeling of phase equilibria and transport properties of refrigerants: free-volume-theory coupled to the soft-SAFT equation of state.** F. Llovell. R.M. Marcos, L.F.Vega. IX Iberoamerican Conference Phase Equilibria and Fluid Properties for Process Design – Equifase 2012. Puerto Varas, Chile, October 8th – 12th, 2012.

40. **Phase, Interfacial, Derivative and Transport Properties of Aqueous Systems with the Soft-SAFT EoS.** O. Vilaseca, F. Llovell and L.F. Vega, AICHE 2012 Annual Meeting, Pittsburgh, USA, October 28th – November 2nd, 2012.

41. **Molecular Simulations of Amine Functionalized MCM-41 for the Separation of CO_2/N_2**. L.F. Vega, S. Builes, Oral Presentation at AICHE 2012 Annual Meeting, Pittsburgh, USA, October 28th – November 2nd, 2012.

42. **Extending the Capabilities of Molecular-Based Equations of State: Simultaneous Calculations of Phase Equilibria and Transport Properties.** F. Llovell; L. F. Vega. 2012 AIChE Meeting, Pittsburgh, USA, October 28th – November 2nd, 2012.

43. **Study of Interfacial Phenomena, Viscosity and Derivative Properties of Common Ionic Liquids with the soft-SAFT EoS**. O. Vilaseca; F. Llovell; M. B. Oliveira; J. A. P. Coutinho; L. F. Vega. 2012 AIChE Meeting, Pittsburgh, USA, October 28th November 2nd, 2012.

44. **Effect of modified atmosphere packaging with argon on the sensorial and microbiological quality of fresh-cut apples**. S. Pardilla, L. F. Vega, M. Mor-Mur, S. Guri. 7th Spanish Congress of Food Engineering. Ciudad Real, Spain, November 2012.

45. **In-situ characterization of PEM membranes by AFM,** I. Torres, A. Arce, J. O. Ossó, L. F. Vega. V National Congress of Fuel Cells - CONAPPICE 2012. Madrid, 21st-23th November 2012.

46. **Development of a new electrochemical process for the synthesis of pharmaceutical products through electrocarboxilation using CO_2**. G. Guirado, I. Reche, A. B. Gómez, I. Gallardo, L. F. Vega, J. O. Ossó. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

47. **Effect of catalytic additives on selectivity and productivity of CO_2 photoreduction using solar radiation.** M. Manzanares, A. Parra, C. Fàbrega, T. Andreu, J. O. Ossó, L. F. Vega, J.R. Morante. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

48. **A new methodology for the evaluation of the CO2 reduction efficiency of TiO_2-based photocatalysts**. C. Roscini, J. Hernando, J. O. Ossó, J. L. Bourdelande, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

49. **New impregnation technique of surface modification based on supercritical CO_2**. P. López-Araunguren, C. Domingo, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

50. **Optimization of new materials for a more efficient CO_2 capture by molecular simulations**. S. Builes, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

51. **Thermodynamic Modeling of the Solubility of supercritical CO_2 and other Gases on Ionic liquids with the soft-SAFT Equation of State.** L. F. Vega, F. Llovell, O. Vilaseca, R. M. Marcos. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

52. **Process for erradication of insect pests in agri-food products**. S. Guri, J. Riudavets, M. J. Pons, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

53. **Synthesis Of Biodiesel Using Supercritical CO_2 As Solvent: Lipids Recovery From Frying Oil.** J. Torres-Sanglas, R. Solanas, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

54. **Acidificación de efluentes de agua dulce procedentes de circuitos de refrigeración de plantas eléctricas. / "Acidification of freshwater effluent from cooling circuits of power plants".** S. Guri, R. F López, E. Rodríguez. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

55. **Degreasing of Tires Powder by Supercritical CO_2.** J. Torres-Sanglas, R. Solanas, J. Godoy, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

56. **Preservation of gluten-free bakery products in modified atmosphere.** S. Guri, J. Riudavets, M. J. Pons, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

57. **Optimization of the treatment with CO_2 of recreational waters (pools)**, W. Chang, A. Quindos, L. F. Vega, S. Guri. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

58. **Preservation of gluten-free bakery products in modified atmosphere**. S. Guri, J. Riudavets, M. J. Pons, L. F. Vega. Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.
59. **Process for erradication of insect pests in agri-food products.** J. Riudavets, L. F. Vega, S. Guri, M. J. Pons (no sé si esta el orden final de los autores). Presentation of the Results of the SOST-CO2 CENIT, Madrid, December 5th, 2012.

2013

60. **Insights into the behavior of organic compounds in water.** F. Llovell. A. García-Sánchez; L.F. Vega. WaterSpain 2013. Zaragoza, Spain, February 7th – 8th, 2013.
61. **Molecular simulations of hybrid materials for CO_2 capture and other gas separation processes.** L.F. Vega, S. Builes. I Workshop of the Molecular Simulation Spanish Network. Baiona, Spain, March 7th – 9th, 2013.
62. **Application of a Renormalization-Group Treatment to SAFT-VR.** E. Forte, F. Llovell, L.F. Vega, J. P. M. Trusler, A. Galindo. PPEPPD 2013, Iguazu Falls, Argentina, May 16th – 21st, 2013
63. **Quantifying the effects of argon and high content of CO_2 on the shelf-life of fresh-cut apples** S. Pardilla, L. F. Vega, M. Mor-Mur, S. Guri. XI International Controlled and Modified Atmosphere Research Conference. Trani, Italy, June 3rd-7th, 2013.
64. **Improving shelf-life by distributing tomatoes in pallets with MAP.** S. Guri, L. F. Vega, L. Romero, M.J. Pons. XI International Controlled and Modified Atmosphere Research Conference. Trani, Italy, June 3rd-7th, 2013.
65. **A combined approach for the modelling of thermophysical properties: molecular simulations and soft-SAFT.** A. García-Sánchez; F. Llovell; D. Bahamón; O. Vilaseca; R. M. Marcos; L. F. Vega XXIX Annual Meeting of Theoretical and Computational Chemistry, Barcelona, Spain, July 10th – 11th, 2013.
66. **Building an accurate and transferable modelling tool for ionic liquids systems.** L.F. Vega, F. Llovell, M. Oliveira, L. Pereira, N. Mac Dowell, J. Coutinho. 7th International Discussion Meeting on Relaxation in Complex Systems. Universitat Politenica de Catalunya, Barcelona, Spain. July 21st – 26th, 2013.
67. **Solubility of pollutants in ionic liquids.** R. M. Marcos; F. Llovell; N. Mac Dowell; L. F. Vega. 7th International Discussion Meeting on Relaxations in Complex Systems, Barcelona, Spain, July 21st – 26th, 2013.
68. **Simulation of materials and processes for energetic applications.** A. García-Sánchez, Marc Castells-Ballesta and L.F. Vega. 7th International Discussion Meeting on Relaxations in Complex Systems, Barcelona, Spain, July 21st – 26th, 2013
69. **The behaviour of aqueous mixtures containing alkanols and ionic liquids: description from soft-SAFT.** F. Llovell; O. Vilaseca; L. F. Vega. 7th International Discussion Meeting on Relaxations in Complex Systems, Barcelona, Spain, July 21st – 26th, 2013.
70. **Development of H_2 Safety Expert Groups and due diligence tools for public awareness and trust in hydrogen technologies and applications.** L.F. Vega, P. Ruiz MATGAS 2000 AIE, Products PLC , European Hydrogen Association, SOLVAY SPECIALTY POLYMERS ITALY S.P.A., Politecnico di Milano, McPhy Energy SA, SOL S.p.A., Ciaotech S.r.l, Technische Universiteit Eindhoven. International Conference on Hydrogen Safety -ICHS 2013, Brussels, Belgium, September, 9th -11th, 2013.
71. **Optimized adsorbent materials for CO_2 capture and separation:advanced experimental techniques and molecular simulations.** L.F. Vega, S. Builes, P. Lopez-Aranguren and C. Domingo. International Symposium on CO2 capture: microscopic studies and applications". University Paris-Est Marne la-Vallee, France. 18th -20th September 2013
72. **Innovative solutions in food preservation: combining biodegradable polymers with map to extend the food shelf-life.** S. Guri, L. F. Vega, M. J. Pons, A. Callens, M. Jackson and F. M. Mimessi. International Nonthermal Food processing Workshop. Florianópolis, Brasil. 30 September-2 October, 2013.

73. **Pest Control in Warehouses through modified atmosphere with CO_2 high levels and addition of SO_2.** M.J. Pons, S. Guri, R. Gabarra, Castañé, C, Alomar. O, Vega, L.F., Ruidavets, J. VIII National Congress of Applied Entomology (SEEA). Tecnocampus Mataró. Barcelona. Spain. October 21st -25th, 2013.

74. **Removing Microorganisms From Wastewater in a Painting Industry By Advanced Oxidation Processes.** W. Chang, L.F. Vega, S. Guri. 2013 AIChE Annual Meeting. San Francisco, USA. November 3-8th, 2013.

75. **Effect of the Impurities On Thermophysical and Transport Properties of CO_2 for Transportation.** F. Llovell, L.F. Vega. AIChE Annual Meeting. San Francisco, USA. November 3-8th, 2013.

76. **Molecular Thermodynamic Characterization of Ionic Liquids for Separation Processes.** L.F. Vega, F. Llovell, M. Oliveira, O. Vilaseca, R.M. Marcos, J. Couthinho. AIChE Annual Meeting. San Francisco, USA. November 3-8th, 2013.

2014

77. **Envasado de cerezas en atmósferas modificadas con biopolímeros como material de envase.** M.J.Pons; L.F. Vega; L. Olivares; S. Guri. 9º Ibero American Congress of Food Engineering (CIBIA9). Universitat Politècnica de Valencia. Valencia, Spain. January 13-16th, 2014.

78. **Viability of an original mix of gases for preservation of fresh tuna.** S. Pardilla, L.F. Vega, S. Guri. 9º Ibero American Congress of Food Engineering (CIBIA9). Universitat Politècnica de Valencia. Valencia, Spain. January 13-16th, 2014.

79. **Modeling biodiesel production and purification: a molecular-based approach with the soft-SAFT EoS.** M. B. Oliveira, F. Llovell, J. A. P. Coutinho, L. F. Vega, 4th International Congress on Green Process Engineering. Sevilla, Spain, April 7-10th, 2014.

80. **Ionic liquids for gas separation of greenhouse and acid gases.** F. Llovell, M. B. Oliveira, J. A. P. Coutinho, R. M. Marcos, L. F. Vega, 4th International Congress on Green Process Engineering. Sevilla, Sevilla, Spain, April 7-10th, 2014.

81. **Optimal Design of the Biodiesel Production and Purification Process: A molecular based approach with soft-SAFT**. M. B. Oliveira, F. Llovell, S. V. D. Freitas, L. F. Vega, J. A. P. Coutinho. SAFT meeting 2014: From SAFT to Experiment and Back. Tróia, Portugal, April 22-24th 2014.

82. **Alternative routes in the synthesis of amino functional sorbents using compressed CO_2.** P. López-Aranguren, L.F. Vega and C. Domingo. 14th European Meeting on Supercritical Fluids. Marseille, France, May 18-21th 2014.

83. **A Statistical Mechanical Equation of State for Critical Property Predictions in Pore Confined Fluids.** P.López-Aranguren, E.H Chimowitz, L. F. Vega, C. Domingo. 14th European Meeting on Supercritical Fluids. Marseille, France, May 18-21th 2014.

84. **Modeling the influence of impurities on supercritical CO_2 pipeline transport with a refined equation of state.** F. Llovell, L. F. Vega, FLUCOMP 2014, Barcelona, Spain, June 10-13th 2014. *Awarded as best talk of the conference.

85. **Alternative routes in the synthesis of amino functional sorbents using compressed CO_2**. P. López-Aranguren, L.F. Vega and C. Domingo. , FLUCOMP 2014, Barcelona, Spain, June 10-13th 2014

86. **High Pressure Separation of Greenhouse Gases Using Different Ionic Liquids Families.** M. B. Oliveira, L. M. C. Pereira, P. J. Carvalho, F. Llovell, L. F. Vega, J. A. P. Coutinho. SAFT meeting 2014: From SAFT to Experiment and Back. Tróia, Portugal, April 22-24th 2014.

87. **Environmental benefits of using Carbon Dioxide for swimming pool water neutralization.** G. Blejman, W. Chang, S. Guri and L.F. Vega. 3rd environmental seminar on: "Subsurface water pollution".. University of Barcelona, Barcelona, Spain. June 26th 2014.

88. **Modeling the Thermophysical Properties of CO_2 Mixtures at Transport Conditions.** F. Llovell, L.F. Vega. 27th European Symposium on Applied Thermodynamics (ESAT 2014): "Experiments meet Theory and Simulation". Eindhoven - The Netherlands. July 6-9th 2014.

89. **Integrated Modeling of Thermodynamic and Transport Properties of Fluids.** F. Llovell, L.F. Vega. 27th European Symposium on Applied Thermodynamics (ESAT 2014): "Experiments meet Theory and Simulation". July 6-9th 2014, Eindhoven - The Netherlands.

90. **Sustainable uses of CO_2**. L.F. Vega, P. Ruiz. Converting CO2 to Chemicals – Horizon 2020 Networking. Aston University, Birmingham, UK. July 8th 2014.

91. **A common framework to model thermodynamic and transport properties: soft-SAFT + Free-Volume Theory.** F. Llovell, L.F. Vega, XIV Encuentro Inter- Bienal del Grupo Especializado de Termodinámica (GET). Baiona, Vigo, 14th-16th September 2014.

92. **Physicochemical characterization of the biodiesel production process using a molecular-based approach.** F. Llovell, M.B. Oliveira, J.A.P. Coutinho, L. F. Vega, 13th Mediterranean Conference on Chemical Engineering (MCCE), Barcelona, Spain, 30th September – 3rd October, 2014.

93. **Separation of Greenhouse Gases using Ionic Liquids with the soft-SAFT EoS.** M. B. Oliveira; L. M. C. Pereira; P. J. Carvalho; F. Llovell; L. F. Vega; J. A. P. Coutinho, Gordon Research Conferences- Green Chemistry, Industrial Successes and Challenges, Hong Kong, China, 27/07/2014-01/08/2014.

94. **Soft-SAFT for the Separation of Greenhouse Gases with Ionic Liquids.** M. B. Oliveira, L. M.C. Pereira, P. J. Carvalho, F.Llovell, L. F. Vega, J. A. P. Coutinho, 20th European Conference on Thermophysical properties (ECTP2014), Porto, Portugal. 31 August- 4 September 2014.

95. **Soft-SAFT for the Optimal Design of the Biodiesel Production and Purification Process.** M. B. Oliveira, F. Llovell, S. V. D. Freitas, L. F. Vega, J. A. P. Coutinho, 20th European Conference on Thermophysical properties (ECTP2014), Porto, Portugal. 31 August- 4 September 2014.

96. **Physicochemical characterization of the biodiesel production process using a molecular-based approach.** F. Llovell, M.B. Oliveira, J.A.P. Coutinho, L. F. Vega, 13th Mediterranean Conference on Chemical Engineering (MCCE), Barcelona, Spain, 30th September – 3rd October 2014.

97. **Viabilidad de la utilización de biopolímeros como material de envase para la conservación de cerezas en atmósfera modificada.** Pons, M.J., Vega, L. F., Olivares, L. and Guri, S. XI Simposio Nacional y VIII Ibérico Sobre Maduración y Postcosecha. Valencia, Spain, October 21 - 23, 2014.

98. **Injection of CO_2 in water: thermophysical and fluid flow modelling using an advanced equation of state and CFD.** F. Llovell, J. Herrmann, W. Chang, S. Guri, L. F. Vega, 13th Mediterranean Conference on Chemical Engineering (MCCE), Barcelona, Spain, 30th September – 3rd October 2014.

99. **Increasing sustainability in energy recovery systems, using Life Cycle Assessment.** G. Blejman, J. Torres, L.F. Vega, SETAC Europe 20th LCA Case Study Symposium, Novi Sad, Serbia, 24-26 November 2014.

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100. **Modeling CO_2 Supercritical Processes Using Molecular-Based Equations with Critical Region Treatments.** F. Llovell, L. F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

101. **Aplicación de Atmósferas Modificadas con alto contenido de CO_2 y biopolímeros para la conservación de cerezas**. Pons, M.J., Vega, L. F., Olivares, L. and Guri, S. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

102. **Evaluation of materials for application in CO_2 capture**. Bahamón, D. and L.F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

103. **Environmental benefits of using Carbon Dioxide for swimming pool water neutralization**. G. Blejman, S. Guri, A. Gomà, W. Chang, J. La Fuente and L.F. Vega. Workshop Aportando Valor al CO2, organized by The

Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

104. **Proyección de la Aplicación Industrial de las Tecnologías con CO_2 Supercrítico**. R. Solanas, J. Torres, D. Quintanilla and L. F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

105. **The SOST-CO2 Project: Bringing New Industrial Sustainable Uses of CO_2**. P. Ruiz, L. F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

106. **Influence of impurities in CO_2 capture over a zeolitic material**. A. Díaz-Márquez, P. Gamallo, D. Bahamón and L. F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

107. **Reciclado de Residuos con CO_2 Supercrítico: Recuperación de Aceite de Freír y Polvo de Neumáticos**. R. Solanas, J. Torres and L. F. Vega. Workshop Aportando Valor al CO2, organized by The Spanish Technological Platform for CO2 (PTECO2) and the Spanish Technological Platform for Sustainable Chemistry (SusChem). Madrid, Spain. February 17-18th, 2015.

108. **Optimal Design of the Biodiesel Production and Purification Process with soft-SAFT**. M. B. Oliveira, F. Llovell, S. V. D. Freitas, L. F. Vega, J. A. P. Coutinho. 41st Conference on Phase Equilibria/ XLIèmes Journées d'Étude des Équilibres entre Phases (JEEP2015). Coimbra, Portugal. March 25-27, 2015.

109. **Separation of Greenhouse Gases with Ionic Liquids with the Soft-SAFT**. M. B. Oliveira, L. M.C. Pereira, P. J. Carvalho, F. Llovell, L. F. Vega, J. A. P. Coutinho. 41st Conference on Phase Equilibria/ XLIèmes Journées d'Étude des Équilibres entre Phases (JEEP2015). Coimbra, Portugal. March 25-27, 2015.

110. **Reproducing the Critical Divergence of the Isochoric Heat Capacity into Soft-SAFT**. F. Llovell, L. F. Vega, M.A. Anisimov, J.V. Sengers. A 25th anniversary celebration of the Statistical Associating Fluid Theory (SAFT). SAFT2015. Houston, USA, May 17-19, 2015.

111. **The extended soft-SAFT equation of state as a tool to assess complex fluids experimental data**. F. Llovell, L. F. Vega. A 25th anniversary celebration of the Statistical Associating Fluid Theory (SAFT). SAFT2015. Houston, USA, May 17-19, 2015.

112. **Testing the Accuracy of the soft-SAFT Parameterization: Simultaneous Description of Phase Equilibria and Derivative Properties**. Mariana B. Oliveira[1], Felix Llovell[2], Lourdes F. Vega[2,3] and João A. P. Coutinho, NINETEENTH SYMPOSIUM ON THERMOPHYSICAL PROPERTIES, Boulder, Colorado, USA. June 21-26, 2015.

113. **Extension of the Soft-SAFT Thermodynamic Model to Derivative and Transport Properties**. F. Llovell, L.F. Vega. X Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE 2015), Alicante, Spain. June 28th-July 1st, 2015.

114. **Modeling the Vapor-Liquid Equilibria and Water Activities of Aqueous Ionic Liquids**. M.B. Oliveira, E. Crespo, F. Llovell, L. F. Vega, J. A. P. Coutinho. X Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE 2015), Alicante, Spain. June 28th-July 1st, 2015.

115. **Phase Equilibria and Self-Aggregation behaviour of Fluorinated Ionic Liquids in Water**. F. Llovell, A. B. Pereiro, J. M. M. Araújo, L. Paulo N. Rebelo, M. M. Piñeiro and L. F. Vega. X Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE 2015), Alicante, Spain. June 28th-July 1st, 2015.

116. **Comparison between different materials for CO_2 capture in a post-combustion stream.** D. Bahamon, L.F. Vega. X Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE 2015), Alicante, Spain. June 28th-July 1st, 2015.

117. **CO_2 separation from multicomponent mixtures by adsorption in MOF Cu-BTC and zeolite 13X.** D. Bahamon, A. Diaz-Marquez, P. Gamallo, L. F. Vega, 6th Iberian Meeting on Colloids and Interfaces Gimerães, Portugal. July 8-10, 2015

118. **Ibuprofen removal from real wastewater combining biological and chemical treatments**. L. Carro, W.Chang, J.A. Torà, A. Bartolí, S. Guri, L.F. Vega. Symposium: S15 Chemical Engineering and Environment. XXXV Bienal RSEQ. La Coruña, Spain. July 19-23, 2015.

119. **Ozone/hydrogen peroxide-based advanced oxidation to remove ibuprofen from aqueous solutions**, W.Chang, L. Carro, S. Guri, L. F. Vega. Symposium: S15 Chemical Engineering and Environment. XXXV Bienal RSEQ. La Coruña, Spain. July 19-23, 2015.

120. **Simultaneous description of phase equilibria and derivative properties for testing the accuracy of the soft-SAFT parameterization.** A. Palma, M.B. Oliveira, F. Llovell, L.F. Vega, J.A.P. Coutinho. Thermodynamics 2015, Copenhaguen, Denmark. September, 15-18, 2015.

121. **soft-SAFT EoS for the vapor-liquid equilibria and water activities of aqueous ionic liquids**. A. Palma, M.B. Oliveira, E. Crespo, F. Llovell, L.F. Vega, J.A.P. Coutinho. Thermodynamics 2015, Copenhaguen, Denmark. September, 15-18, 2015.

122. **Computational evaluation of different materials for CO_2 capture**. D. Bahamon, L.F. Vega. IV New trends in Computational Chemistry for Industry Applications. Barcelona, Spain. October 1-2, 2015.

123. **Characterization of the biodiesel production process using a generalized predictive approach with a molecular-based equation of state**. F. Llovell, M.B. Oliveira, J.A.P. Coutinho, L. F. Vega. IV New trends in Computational Chemistry for Industry Applications, Barcelona, Spain. October 1-2, 2015.

124. **Vapor-liquid equilibria and water activities of aqueous solutions of ionic liquids with the soft-SAFT EOS.** E. Crespo, M. B. Oliveira, F. Llovell, L. F. Vega, J. A. P. Coutinho, VIII Congresso Brasileiro de Termodinâmica, CBTermo 2015, Aracaju-Sergipe, November 16-20 , 2015.

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125. **Simultaneous description of second order thermodynamic derivative properties and phase equilibria by soft-SAFT: evaluation and improvements**. M.B. Oliveira, F. Llovell, J.A.P. Coutinho, L.F. Vega, 14th PPEPPD, International Conference on Properties and Phase Equilibria for Product and Process Design, Porto, 22-26 May 2016.

126. **Fine-Tuning of Phase and Self-Aggregation Behaviour of Fluorinated Ionic Liquids in Aqueous Solutions**. A.B. Pereiro, F. Llovell, J. M. M. Araújo, L. P. N. Rebelo,M.M. Pineiro L. F. Vega, 14th PPEPPD, International Conference on Properties and Phase Equilibria for Product and Process Design, Porto, 22-26 May 2016.

127. **A molecular model to characterize the effect of impurities on the thermophysical properties on CO_2 transport**. F. Llovell, L. F. Vega, 14th PPEPPD, International Conference on Properties and Phase Equilibria for Product and Process Design, Porto, 22-26 May 2016.

128. **Computational assessment between different-type materials for CO2 capture**. D. Bahamon, L. F. Vega, 14th PPEPPD, International Conference on Properties and Phase Equilibria for Product and Process Design, Porto, 22-26 May 2016.

129. **Solubility of respiratory gases in fluorinated ionic liquids**. A.B. Pereiro, M. L. Ferreira, J. M. M. Araújo, L. P. N. Rebelo, L. F. Vega, and F. Llovell, EUCHEM 2016, 16th EUCHEM on Molten Salts and Ionic Liquids, Vienna, Austria , July 3 - 8, 2016

130. **Comparative study of MOFs and zeolites for CO2 capture and separation at process conditions**. L.F. Vega, D. Bahamon. Abu Dhabi International Petroleum Exhibition and Confrence, ADIPEC2016, Abu Dhabi, UAE, November 7-10, 2016.

131. **Modeling Gas Separation of Greenhouse Gases with Ionic Liquids and Deep Eutectic Solvents**. F. Llovell, J. Orduña, L. F. Vega. AIChE Annual Meeting. AICHE2016. San Francisco, California, USA, November 13-18, 2016.

132. **Physico-chemical Characterization of Fluorinated Ionic Liquids for their Use as Blood Substitutes.** F. Llovell, M. Ferreira, A. B. Pereiro, J. M. M. Araújo, M. M. Piñeiro, L. P. N. Rebelo, L. F. Vega. AIChE Annual Meeting. AIChE2016. San Francisco, California, USA, November 13-18, 2016.

133. **Comparative study of MOFs and zeolites for CO2 capture and separation at process conditions**. L.F. Vega, D. Bahamon. AIChE Annual Meeting. AICHE2016. San Francisco, California, USA, November 13-18, 2016.

134. **Development of a Predictive EoS for Oil Characterization, Including Asphaltene Behavior**. W. A. Fouad, S. M. AlHassan, L. F. Vega. PIRC R&D Conference and Exhibition, The Petroleum Institute, Abu Dhabi, UAE, November 21st, 2016.

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135. **New Artificial Oxygen Carriers using Fluorinated Ionic Liquids.** M. L. Ferreira, F. Llovell, K. Shimizu, J. M. M. Araújo, J. N. C. Lopes, L. P. N. Rebelo, M. M. Piñeiro, L. F. Vega, A. B. Pereiro. ILWS - Winter School on Ionic Liquids. University of Porto, Porto, Portugal, 7-10th February, 2017

136. **Modeling Pharmaceuticals Removal from Water Effluents by Adsorption in Activated Carbons.** L.F. Vega, D. Bahamon. The Fith International Conference on Water, Energy and Environment (ICWEE/5). American University of Sharjah, UAE, February 28-March 2, 2017.

137. **Viscosity modeling of n-alkanes and linear polyethylene with soft-SAFT + Free Volume Theory**. E. Cane, F. Llovell, L. F. Vega. 2017 PIRC R&D Conference and Exhibition, The Petroleum Institute, Abu Dhabi, UAE, March 19-20, 2017.

138. **Thermodynamic Characterization of Fluorinated Ionic Liquids to be used as Artificial Oxygen Carriers**. A.B. Pereiro, M. L. Ferreira, F. Llovell, J. M. M. Araújo, L. P. N. Rebelo, M. M. Piñeiro, L. F. Vega, 16th European Meeting on Supercritical Fluids (EMSF), Lisbon, Portugal, April 25-28, 2017.

139. **Studying the Aneotropic Behavior of Azeotropic Refrigerants Using Molecular Theory and Simulation**. W.A. Fouad, L.F. Vega, 2017 SAFT Conference, Heidelberg, Germany, May 11-13, 2017.

140. **Development of transferable models to describe the physicochemical behavior of ionic liquids and deep eutectic solvents with a molecular-based equation**. J.O. Lloret, L.F. Vega, F. Llovell, XXXIII Annual meeting of the Reference Network of R+D+I on Theoretical and Computational Chemistry, Tarragona, Spain, July 5-6th, 2017.

141. **Accurate modeling of interfacial and phase equilibrium properties of reservoir fluids.** L. M.C. Pereira, L.F. Vega, 21st European Conference on Thermophysical Properties, ECTP 21st, Graz, Austria, September 3 – 8, 2017.

142. **CO2 capture with glymes - soft-SAFT EoS modeling**. E.A. Crespo, J.M. L. Costa, M. Amaral, C. Dariva, P. Carvalho, L. F. Vega, J.A. P. Coutinho. Thermodynamics 2017. Edinburgh, UK, September 5-8, 2017.

143. **Development of transferable molecular models of conventional MEA and alternative Deep Eutectic Solvents for CO2 capture with soft-SAFT.** J.O. Lloret, L.F. Vega, F. Llovell. Thermodynamics 2017. Edinburgh, UK, September 5-8, 2017.

144. **Investigation of interfacial properties in multiphase carbon dioxide + oil/natural gas + water systems using the DGT + soft-SAFT approach**. L. Pereira, F. Llovell, J. Torné, L.F. Vega. World Conference on Chemical Engineering (WCCE10), Barcelona, Spain, October 1-5, 2017

145. **An Accurate Molecular-Based Equation of State for the Simultaneous description of Thermodynamic and Transport Properties of Complex Mixtures of Industrial Interest**. F. Llovell, J. O. Lloret, E. Cané, L. F. Vega. World Conference on Chemical Engineering (WCCE10), Barcelona, Spain, October 1-5, 2017

146. **Pharmaceuticals Removal from Water Effluents by Adsorption in Activated Carbons Using Monte Carlo Simulations.** Daniel Bahamon and Lourdes F. Vega. ESCAPE27, World Conference on Chemical Engineering (WCCE10), Barcelona, Spain, October 1-5, 2017.

147. **Thermodynamic Modeling of Alternative Solvents for CO2 Capture with the Soft-SAFT Equation of State** J. O. Lloret, F. Llovell, L. F. Vega. World Conference on Chemical Engineering (WCCE10), Barcelona, Spain, October 1-5, 2017.

148. **Effect of amine density on the CO2 separation properties of functionalized silica materials.** S. Builes, P. López- Aranguren, J. Fraile, C. Domingo, L. F. Vega. World Conference on Chemical Engineering (WCCE10), Barcelona, Spain, October 1-5, 2017.

149. **Development of a new and transferable molecular model for an accurate thermodynamic description of Deep Eutectic Solvents with soft-SAFT**. O. Lloret, L. F. Vega, F. Llovell. IV International Conference on Ionic Liquid-Based Materials (IMAT), Santiago de Compostela, Spain, October 24-27, 2017.

150. **Theoretical-experimental Characterization of Fluorinated Ionic Liquids.** M. L. Ferreira, F. Llovell, J.M.M. Araújo, L.P.N. Rebelo, L. F.Vega, A.B. Pereiro. IV International Conference on Ionic Liquid-Based Materials (IMAT), Santiago de Compostela, Spain, October 24-27, 2017.

151. **CO2 Capture With Glymes – Soft-Saft EOS Modeling**, K. Campos, M.J. Amaral, E. Crespo, C. Dariva, L.F. Vega, P. Carvalho, JAP Coutinho, IX CBTERMO, Porto Alegre, Brazil, October 23-27, 2017

152. **Understanding the Role of Intermolecular Forces in Molecular-Based Equations of State: 20 Years of the Soft-SAFT Equation**. L.F. Vega, F. Llovell, F.J. Blas, O. Vilaseca, J. C. Pàmies, JAP Coutinho, J. Torné, AIChE Annual Meeting, Minneapolis, MI, USA, 29 October- November 3rd, 2017.

153. **Molecular Dynamic Simulations of Liquid/Vapor Interface for Hydrofluoroolefine (HFO) working fluids**. Y. Li, W.A. Fouad, L.F. Vega, Environmental & Materials Science Symposium, Khalifa University of Science and Technology, December 6, 2017.

2018

154. **Molecular Simulations of Nature-inspired Hierarchical Porous Materials for Energy Storage**. M. Elabyouki, L.F. Vega, M. Khaleel, Fourth UAE Graduate Students Research Conference (GSRC 2018), American University of Sharjah, Sharjah, UAE, April 21, 2018.

155. **Molecular Dynamic Simulations of Low Global Warming Potential Refrigeran**ts. Y. Li, W.A. Fouad, L.F. Vega, Fourth UAE Graduate Students Research Conference (GSRC 2018), American University of Sharjah, Sharjah, UAE, April 21, 2018.

156. **Characterization of Reservoir Fluids: A Predictive Model for Interfacial and Bulk Phase Equilibrium Properties**. L.M.C. Pereira, L.F. Vega, A. Chapoy, B. Tohidi, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

157. **Accurate Viscosity Predictions of Carbonated Aqueous MEA Solutions for CO_2 capture processes**. L.M.C. Pereira, L.F. Vega, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

158. **Development of predictive molecular models to describe thermodynamic and transport properties of hydrocarbons and their mixtures present in petroleum fractions**. E. Cane, F. Llovell, L.F. Vega, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

159. **Exploring alternative solvents for gas processing using the soft-SAFT EoS**. E.A. Crespo, P.J. Carvalho, J.A. P. Coutinho, L.F. Vega, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

160. **Development of a Predictive Molecular Model for Abu Dhabi Crude Oils Phase Behavior**. W.A. Fouad, M.I. L. Abutaqiyab, K. Mogensen, Y. F. Yap, A. Goharzadeh, F.M. Vargas, L.F. Vega, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

161. **Amine-Functionalized Hierarchical Zeolites for Carbon Dioxide Capture**. M. Khaleel, A. Tuneu Pujolras, G. Karanikolos, L.F. Vega, F. Llovell, RDPetro Conference, Abu Dhabi, UAE, May 9-10, 2018.

162. **A coarse-grained model of choline chloride for the modelling of deep eutectic solvents using the soft-SAFT equation of state**. E.A. Crespo, L. P. Silva, J. O. Lloret, P. J. Carvalho, L. F. Vega, F. Llovell, J. A. P. Coutinho. ESAT, Prague, June 10-13, 2018.

163. **A soft-SAFT Robust Model for the Thermodynamic Characterization of CO_2 Absorption in Aqueous Mixtures of Amines**. L.M.C. Pereira, F. Llovell, L.F. Vega. The Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29, 2018.

164. **Accurate Modeling of Interfacial and Phase Equilibrium Properties of Reservoir Fluids with the soft-SAFT + DGT Approach**. L.M.C. Pereira, L.F. Vega. The Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29, 2018.

165. **Soft-SAFT Transferable Molecular Models for the Description of Gas Solubilities in Eutectic Ammonium Salt Solvents**. R.M. Ojeda, L.F. Vega, F. Llovell. The Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29, 2018.

166. **Application of Molecular Modeling Tools for the Rationale Design of New Low Global Warming Potential Refrigerants**. W.A. Fouad, Y. -T Li, L.F. Vega. The Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29, 2018.

167. **Physicochemical Characterization of Fluorinated Ionic Liquids for Biomedical Applications**. F. Llovell, A.B. Pereiro, M. L. Ferreira, R. M. Ojeda, J.M. M. Araújo, L. P. N. Rebelo, M. M. Piñeiro, L.F. Vega. The Twentieth Symposium on Thermophysical Properties, Boulder, CO, USA, June 24-29, 2018.

168. **Design and optimization of processes with novel adsorbent materials for CO_2 capture by a combined molecular simulations-experimental approach**. D. Bahamon, S. Builes, M. Abu Zahra, L. F. Vega. 14th International Conference on Greenhouse Gas Control Technologies, GHGT-14, Melbourne, Australia, 21st -25th October 2018.

169. **Integrated Approach for CO_2 Capture Processes: Low pressure CO_2 capture from SRU Tail Gas**. L. M. C. Pereira, P. Brandl, M. T. Mota-Martinez, N. Mac Dowell, L. F. Vega. 14th International Conference on Greenhouse Gas Control Technologies, GHGT-14, Melbourne, Australia, 21st -25th October 2018.

170. **Towards a Solvent Screening Tool for CO_2 Capture Applications: Predicting the Phase Behaviour of CO2-Amine Blends from Single Amine Mixtures**, L.M.C. Pereira, L.F. Vega, XI Iberoamerican Conference on Phase Equilibria and Fluid Properties for Process Design (EQUIFASE), Cordoba, Argentina, 22-25 October 2018.

171. **Thermophysical, Interfacial and Transport Properties of Low Global Warming Potential Refrigerants from Molecular Theory and Simulations**, W.A. Fouad, Y. Li, L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2018, Pittsburgh, PA, USA, October 28 – November 2, 2018

172. **Amine-Functionalized Hierarchical Zeolites for Carbon Dioxide Capture**, M. Khaleel, A. Tuneu-Pujolras, I. Ismail, G.N. Karanikolos, L. F. Vega, F. Llovell, American Institute of Chemical Engineers Annual Meeting, AIChE2018, Pittsburgh, PA, USA, October 28 – November 2, 2018

173. **Aerated Concrete Blocks Production Using Secondary Steelmaking Slag**, N. Alamoodi, S. Khalil, H. A. Alblooshi, A.J. Alzaabi, M.S. Alqemzi, M. Shahtout, S. Shivaswamy, L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2018, Pittsburgh, PA, USA, October 28 – November 2, 2018

174. **Towards the Development of a Solvent Screening Tool for CO_2 Capture Using Molecular Thermodynamics**, L. M.C. Pereira, F. Llovell, I. I. Alkhatib, L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2018, Pittsburgh, PA, USA, October 28 – November 2, 2018

175. **Adsorption of gases in Metal Organic Frameworks (MOFs) by molecular simulations**, D. Bahamon, S. Builes, L. F. Vega, 2nd Middle Estern Materials Science Conference, NYUAD, Abu Dhabi, UAE, 12-14 November 2018

176. **Molecular simulation of pharmaceutical removal from water by adsorption using activated carbons**, D. Bahamon, A. E. Ogungbenro, M.R.M. Abu Zahra, L. F. Vega, 2nd Middle Eastern Materials Science Conference, NYUAD, Abu Dhabi, UAE, 12-14 November 2018

177. **Tuning adsorbent materials to optimize CO_2 capture at industrial conditions**. D. Bahamon, L. F. Vega, 2nd International Conference on Functional Materials and Chemical Engineering, ICFMCE, Khalifa University, Abu Dhabi, UAE, 20-22 November 2018.

178. **Water for the Future: Hybrid Graphene Nanostructure for Sustainable Water Treatment**. E.S. Cho, D. Bahamon, L.F. Vega, KAIST-KU International Workshop, Abu Dhabi, 6th December 2018.

179. **Molecular modeling as a useful tool for the *ad-hoc* design of adsorbent and catalyst materials,** L. F. Vega, **"**Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

180. **Process optimization using novel adsorbent materials for CO_2 capture at industrial conditions: a molecular simulation approach,** D. Bahamon, M. Abu Zahra, S. Builes, L. F. Vega, **"**Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

181. **Modeling activated carbons for removal of pharmaceutical compounds from water,** D. Bahamon, L. F. Vega, **"**Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

182. **Simultaneous Absorption of H_2S and CO_2 in Aqueous Amine Solvents Using Molecular-Based Models,** I.I. Alkhatib, L.M.C. Pereira, L. F. Vega, Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

183. **Molecular Models for Thermodynamic Properties and Process Modeling of CO_2 Capture with Aqueous Amines,** L.M.C. Pereira, L. F. Vega, Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

184. **Understanding the corrosion inhibitor film on metal surfaces by experiments and molecular simulations,** H. Nasim, L. F. Vega, R. Nogueira, Catalysis and Separation: Shaping the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

185. **Influence of the fluorinated ionic liquids structural features in their thermodynamic and solubility behaviour,** M. L. Ferreira, J. M. M. Araújo, A.B. Pereiro, L. F. Vega, "Catalysis and Separation: Shap ing the Future Energy" workshop CeCas Center, Khalifa University, Abu Dhabi, UAE, 9-10th December 2018.

2019

186. **Understanding the adsorption behavior of corrosion inhibitors on metal surface through molecular simulations**. S. Alareeqi, D. Bahamon, R. Nogueira, L. F. Vega, Fifth UAE Graduate Students Research Conference (GSRC 2019), Abu Dhabi, UAE, April 16th, 2019.

187. **Assessing the use of ionic liquids as a novel solvent platform for the selective recovery of F-gases with the soft-SAFT EoS**. L.F. Vega, C. Albà, S. Asensio, G. Zarca, A. Urtiaga, F. Llovell, International Conference on Phase Equilibria Properties for Process and Product Design (PPEPPD2019), University of British Columbia, Vancouver, Canada, May 12-16th 2019.

188. **Molecular Modelling Approach on the Thermophysical Properties of Fluorinated Ionic Liquids: Influence of the Molecular Structure**. M. Ferreira, L.F. Vega, J. M. M. Araujo, A.B. Pereiro, International Conference on Phase Equilibria Properties for Process and Product Design (PPEPPD2019), University of British Columbia, Vancouver, Canada, May 12-16th 2019.

189. **Molecular simulations of hybrid graphene membranes for CO_2 capture and water treatment**. D. Bahamon, K. R. Bang, E. S. Cho, M. Khaleel, L. F. Vega, International Conference on Phase Equilibria Properties for Process and Product Design (PPEPPD2019), University of British Columbia, Vancouver, Canada, May 12-16th 2019.

190. Describing the impact of impurities on the thermophysical properties of CO_2 at pipeline transport conditions with the extended soft-SAFT EoS. F. Llovell, L.F. Vega. 3rd International Conference on Chemical Engineering (ANQUE-ICCE 3), Santander, Spain, June 19-21st, 2019. ISBN: 978-84-09-12430-5, page 369.

191. Modelling the thermophysical properties and phase equilibria of deep eutectic solvents. E. A. Crespo, L. P. Silva, F. Llovell, P.J. Carvalho, L. F. Vega, J. A. P. Coutinho, 3rd International Conference on Chemical Engineering (ANQUE-ICCE 3), Santander, Spain, June 19-21st, 2019. . ISBN: 978-84-09-12430-5, page 371

192. Theoretical Characterization of 4th Generation Refrigerants based on HydroFluoroOlefins with the extended soft-SAFT EoS, C. Albà-Garriga, D. Jovell, L. F. Vega, F. Llovell. 3rd International Conference on Chemical Engineering (ANQUE-ICCE 3), Santander, Spain, June 19-21st, 2019.ISBN: 978-84-09-12430-5, page 384.

193. CO_2 capture on activated carbons: from experiments to simulations, D. Bahamon, A. E. Ogungbenro, M. Khaleel, M. Abu-Zahra and L. F. Vega**,** *Iberoamerican Symposium on Materials by Design: From the Nano-scale to Engineered Materials* (ANQUE-ICCE3-CIBIQ), Santander, Spain, June 19-21st, 2019. ISBN: 978-84-09-12431-2, page 247.

194. Integrated approach for the design of novel adsorption processes for CO2 capture using multiscale modelling, A. Alhajaj, S. Almenhali, H. Balogun, D. Bahamon, L. F. Vega, IEAGHGG 5th Post-Combustion Capture Conference, PCCC-5, Kyoto, Japan, Sep 17-19, 2019

195. A Robust Molecular-Based Screening Tool for New Solvents for Post-Combustion CO2 Capture, I. I. I. Alkhatib, L. M.C. Pereira, A. Al Hajaj, M. Abu Zahra, L. F. Vega, IEAGHGG 5th Post-Combustion Capture Conference, PCCC-5, Kyoto, Japan, Sep 17-19, 2019

196. CO_2 capture using biomass- derived activated carbons. A. Ogungbenro, L.F. Vega, M. Abu Zahra, IEAGHGG 5th Post combustion Capture Conference (PCCC-5), Kyoto, Sep 2019

197. Systematic Evaluation of Thermophysical Properties of Low GWP Refrigerants and Blends for Refrigeration Cooling Systems By Molecular Modeling Techniques, C. Albà-Garriga, W. A. Fouad, F. Llovell and L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

198. Understanding the Behavior of Degraded Amine Products on CO2 Capture in Aqueous Amines By Molecular Simulations, C.G. Martinez, D. Bahamon, F. Llovell, L.F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

199. **Molecular Simulations of Carbon-Based Adsorbents/Membrane Materials for Sustainable Processes,** D. Bahamon, N. Alkhatib, K. R. Bang, E. S. Cho, M. Abu Zahra, M. Khaleel and L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

200. **New Insights into the Effect of CO_2 and NaCl Aqueous Solution on the Adsorption Behavior of Corrosion Inhibitors through Molecular Simulations**, D. Bahamon, S. Alareeqi, H. Nasim, R. Nogueira, L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

201. **Understanding the Adsorption of 4th Generation Refrigerants on Metal-Organic Frameworks by Molecular Simulations**, D. Bahamon, H. Al Araj, A. Al Najjar, E. J. Garcia, TJ Zhang, M. Khaleel, L. F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

202. **A Molecular-Based Screening Tool for Water-Free Amine Solvents for CO2 Capture**, I.I.I. Alkhatib, L.M.C. Pereira, A. Al Hajaj, M. Abu Zahra, L.F. Vega, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

203. **Complete Surface Tension Characterization of Fluorinated Alcohols and Their Mixtures with Hydrogenated Alcohols: Experimental, Soft-SAFT-DGT Modeling and MD Simulations**, J. Justino, G. M. C. Silva, P. Morgado, L. M.C. Pereira, L. F. Vega, E. Filipe, American Institute of Chemical Engineers Annual Meeting, AIChE2019, Orlando, FL, USA, November 10-15, 2019.

204. **Design of hybrid graphene oxide membranes via crown ether intercalation for water desalination**, D. Bahamon, E. S. Cho, K. R. Bang, L. F. Vega, International Conference on Nanomaterials (e-conference), Sofia,. Bulgary (on-line), May 19-20, 2020.

RESEARCH PROJECTS

EXECUTIVE SUMMARY: FUND RAISING AND TOPICS

Chair Professor and Executive Senior Technical Director with proven experience on leading large R&D projects, moving them from ideas to well defined commercial products, or solving challenging scientific/technical problems.
Lead of 3 **multiannual large public-private consortia**. Managed budget approx. 30MM€ (value 100MM€).
Participation in 13 **EU funded projects**. **Leader (Coordinator)** of 1 of them (9 partners, 1.2MM Euros).
Participation as a **Scientific Lead** in 50+ competitive research grants (EU, USA, Spain, Middle East), **Principal Researcher** of 40 of them. R&D Group recognized as a **Group of Excellence** by external evaluators.
60+ R&D projects with **international companies**.

Main areas of expertise: molecular modelling, clean energy, CO_2 capture and utilization, oil and gas, hydrogen, nanotechnology, low GWP refrigerants, water treatment.

Enclosed is a **detailed list of representative externally funded projects (non-confidential) and industrial projects (title orientative, confidential)** in which I have been involved in the fund raising, technical work and/or management and leadership.

NEW AND CURRENT PROJECTS UNDER EXECUTION (2019-) – TOTAL BUDGET: $19.132 MM (TBU)

1.	**CO_2 capture, energy**	**Title:** *Prime Solvent candidates for next generation of PCC plants* **Type of contract:** R&D project **Funding institution/company:** International Energy Agency (IEAGHG), IEA/CON/20/265 **Participants:** Khalifa University and IEAGHG. **Period:** 3/ 2020-12/2020 **Principal Investigator:** Lourdes F. Vega. **Participants:** M. Abu Zahra, A. Al Hajaj, D. Bahamon, I. Alkhatib **Budget**: AED 190.000 (40.000£)
2.	**Materials, CO_2, H_2**	**Title:** *Research and Innovation Center on CO_2 and H_2 [RICH]* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology, RCII-2019-007 **Participants:** Khalifa University. Supported by 20 academic, industrial and governmental organizations. **Period:** 7/ 2019-6/2022 **Principal Investigator:** Lourdes F. Vega. **Participants:** 15 faculty from KU, 40+ researchers **Budget**: AED 5.516 MM ($ 1.5MM) (excluding CAPEX)
3.	**Materials, Sustainability**	**Title:** *From Waste to Bio-refinery: Renewable Transportation Fuels from Bio-oil Catalytic Upgrading* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology (CIRA project) **Participants:** Khalifa University of Sicence and Technology **Period:** 7/ 2020-12/2023 **Principal Investigator:** K. Polychronopoulou. Role: Co-P. Investigator **Participants**: 3 faculty, 2 researcher, 2 students **Budget**: AED 2.760 MM ($ 0.751MM)
4.	**Materials, CO_2**	**Title:** *Integrated approach to optimize low pressure CO2 capture processes for Enhanced Oil Recovery [IA-CO2] (Phase I and II)*

		Type of contract: R&D **Funding institution/company:** GRC/ADNOC Gas Processing, Total, Shell, Partex **Participants:** Khalifa University (the Petroleum Institute- PI), Imperial College of London (IC), UK **Period:** 1/ 2017-9/2020 **Principal Investigator:** Lourdes F. Vega. **Participants:** 3 from KU/PI, 1 from KU/Masdar Institute, 2 from IC **Budget:** AED 5.281 MM ($ 1.44 MM). [Phase I: AED 1.926 MM ($ 0.525MM). Phase II: AED 3.355 MM ($ 0.915MM)]
5.	**Materials, energy**	**Title:** *Novel Materials for Clean Energy Applications* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology (CIRA project) **Participants:** Khalifa University of Sicence and Technology **Period:** 7/ 2018-12/2021 **Principal Investigator:** Maryam Khaleel. Role: Investigator **Participants:** 2 from PI/KU, 2 from Masdar Institute/KU **Budget**: AED 2.760 MM ($ 0.751MM)
6.	**Materials, separation**	**Title:** *Understanding and optimizing the performance adsorbents and catalysts from molecular modeling* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology (RCII2018-024, CeCaS) **Participants:** Khalifa University of Sicence and Technology **Period:** 7/ 2018-12/2021 **Principal Investigator:** Kyriaki Polychronopoulou. Role: P Investigator of the project **Participants:** 15 faculty, 30+ researchers **Budget:** AED 3.308 MM ($0.9MM)
7.	**Oil & Gas Characterization**	**Title:** *Strategies for Monitoring, Prediction and Remediation of Asphaltene Deposition* **Type of contract:** R&D **Funding institution/company:** ADNOC **Participants:** Khalifa University (the Petroleum Institute- PI), Rice University (RU), USA **Period:** 6/2014-6/2019 **Project Principal Investigator:** Afshin Goharzadeh. Co-I: Lourdes F. Vega in Molecular Modeling Characterization **Participants:** 5 from PI, 4 from Rice University **Budget**: $ 5.18 MM
8.	**Materials, oil and gas**	**Title:** *Behavior of corrosion inhibitors films in specific operating conditions* **Type of contract:** R&D (GRC2017-005) **Funding institution/company:** GRC/ADNOC Gas Processing, Total, Shell, Partex **Participants:** Khalifa University (the Petroleum Institute- PI), TOTAL **Period:** 7/ 2018-6/2020 **Principal Investigator:** Ricardo Nogueira. Co-PI: Lourdes F. Vega in molecular modeling **Participants:** 3 from KU, 2 from TOTAL **Budget**: AED 3.963 MM ($ 1.08MM)
9.	**Water treatment**	**Title:** *Water for the Future: Hybrid Graphene Nanostructure for Sustainable Water Treatment* **Type of contract:** R&D **Funding institution/company:** 2018 KKI Seed Money Project (KU-KAIST) **Participants:** KAIST, Khalifa University of Science and Technology **Period:** 3/2018-3/2020 **Principal Investigator:** Eun Seon Cho (KAIST, Korea). Co-PI: Lourdes F. Vega (KU, UAE)

	Participants: 2 from KU, 2 KAIST
10. Water treatment	**Title:** *Nano Surface Engineered Janus magnetic Particles for Oil Produced Wate Treatment* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology (CIRA2019-007) **Participants:** Khalifa University of Sicence and Technology **Period:** 7/ 2019-12/2022 **Principal Investigator:** Linda Zou. Role: co-PI: Lourdes Vega, Kean Wang **Participants:** 3 faculty, 2 Research Scientist, 2 students **Budget**: AED 2.745 MM ($ 0.748MM)
11. Sustainability, refrigerants	**Title:** *Reduction of the Environmental Impact of Fluorinated Gases in the Sudoe space through Essential Facilitating Technologies* **Type of contract:** R&D **Funding institution/company:** SUDOE, European Union **Participants:** 14 partners and 3 associate partners (one of them PI, part of Khalif University) **Period:** 6/2018-3/2021 **Principal Investigator:** Ana Pereiro (U. Lisbon, Portugal). PI representing associate partner PI/KU: Lourdes F. Vega **Participants:** more than 40 PhDs from the 14 partners and 3 associate partners **Budget**: € 1,742,800 ($1.952MM)
12. Sustainability, refrigerants, materials	**Title:** *Environmentally Friendly Refrigeration Cooling* **Type of contract:** R&D **Funding institution/company:** Khalifa University of Science and Technology (CIRA project) **Participants:** Khalifa University of Sicence and Technology **Period:** 7/ 2018-12/2021 **Principal Investigator:** TJ Zhang. Co- PI: Lourdes F. Vega **Participants:** 3 from PI, 2 from Masdar Institute **Budget**: AED 2.76 MM ($ 0.751MM)
13. Modeling, nanotechnology	**Title:** *Theoretical Chemistry and Computational Modeling - TCCM* **Type of contract:** R&D (610588-EPP-1-2019-1-ES-EPPKA1-JMD-MOB) **Funding institution/company:** European Union. Erasmus Mundus Joint Master Degree (EMJMD) **Participants:** Khalifa University of Sicence and Technology **Period:** 9/ 2019-8/2023 **Principal Investigator:** Manuel Alcami, Universidad Autonoma Madrid. PI from KU: Lourdes F. Vega **Participants:** 9 partners and 33 associated partners. **Budget**: € 4.424 MM ($ 5MM)
14. Sustainability, Materials	**Title:** *High Quality Hydrocarbon Fuels through Catalytic Hydrodeoxygenation (HDO) Technology* **Type of contract: R&D** **Funding institution/company:** Abu Dhabi Award for Research Excellence (AARE) – 2019 **Participants**: Khalifa University of Sicence and Technology **Period:** 1/ 2020-12/2021 (2 years) **Principal Investigator:** K. Polychronopoulou, Co- PI: Lourdes F. Vega, M. Khaleel **Participants:** **Budget**: 979,702 AED

OTHER REPRESENTATIVE PROJECTS ALREADY FINISHED, CLASSIFIED BY TYPE, TOPIC AND GEOGRAPHICAL AREAS

LARGE PUBLIC-PRIVATE R&D CONSORTIA (GEOGRAPHICAL AREA: SPAIN/EU)

1. CO_2 capture & applications, Energy, Food, Water, Sustainability	**Title**: *New sustainable industrial uses of CO_2 - CENIT SOST-CO2* **Coordinator**: Lourdes F. Vega - Carburos Metálicos/Air Products **Funding entity**: Spanish Government CENIT Program- Ministerio de Innovación, Centro para el Desarrollo Tecnológico Industrial (CDTI) and the industrial partners. **Start-End date**: 2008-2011. Period: 4 years **Participating entities**: Industrial partners: Carburos Metálicos/Air Products (Coordinator), Iberdrola Generación, Ros Roca Indox Cryo Energy SL, Linpac Packaging, Nutreco, Repsol, Abengoa Bioenergía (San Roque; Bioetanol Galicia), Inabensa, Aqualogy, Idesa, Amphos21, BioGas Fuel Cells, Ultrasen. Collaborators: 31 public research institutions. **Budget:** 26.4 MM€
2. Alternative energy, Water treatment, Sustainability	**Title:** *Sustainable Biorefineries – CENIT BIOSOS* **Coordinator**: Ricardo Arjona, Abengoa Bioenergía Nuevas Tecnologías (ABNT) **Principal Investigator (PI)**: Lourdes F. Vega – Carburos Metálicos/Air Products **Funding entity**: Spanish Government CENIT Program- Ministerio de Innovación, Centro para el Desarrollo Tecnológico Industrial (CDTI) and the industrial partners. **Start-End date**: 2009-2012. Period: 4 years **Participanting entities**: Industrial partners: Abengoa Bioenergia Nuevas Tecnologias (Coordinator), Acciona, Azvi, Biopolis, Carburos Metálicos/Air Products, Gairesa, Guascor, Industrias Omar, INFOSA, Krafft, Neuron, Sniace, Solintel, Solutex, Técnicas Reunidas. Collaborators: 31 public research institutions **Budget:** 27 MM€
3. Water treatment	**Title**: Development and integration of biological and advanced oxidation treatments for the sustainable reduction of microcontaminants in wastewater. **Coordinator**: Lourdes F. Vega, Carburos Metálicos/Air Products **Funding entity**: Catalan Government and FEDER – NUCLI program **Start-End date:** 11/02/2013 – 04/05/2015 **Participating entities**: Companies: Carburos Metálicos/Air Products (Coordinator), AERIS, APLICAT, HIDRONIT. R&D Institutions: MATGAS 2000 AIE, BIO-GLS, AMIC. **Budget:** 0.8 MM€

EXTERNAL COMPETITIVE FUNDING - EUROPEAN FUNDED PROJECTS (GEOGRAPHICAL AREA: EUROPE)

1. Ionic iquids, Modeling	**Title:** *EXIL- Exchange on Ionic Liquids* **Coordinator:** Prof Rasmus Fehrmann, Technical University of Denmark **Principal Investigator**: Lourdes F. Vega, MATGAS 2000 AIE **Funding entity:** European Union **Programme:** Chemistry and Molecular Sciences and Technologies (CMST) COST Action CM1206 **End of Action:** 14/05/2017 **Participating entities:** 170 researchers coming from 24 Countries **Budget:** 92 MM€
2. Hydrogen,	**Title:** *Development of H_2 Safety Expert Groups and due diligence tools for public*

Sustainability	*awareness and trust in hydrogen technologies and applications – H_2TRUST*. **Coordinator**: Lourdes F. Vega, MATGAS 2000 AIE, Spain. **Funding entity:** European Union, Fuel Cells and Hydrogen Join Undertaking **Programme:** FCH-JU-2012-1 (topic: SP1-JTI-FCH.2012.5.5) (Support Action Project) **Start-End date:** 01/06/2013 - 28/02/2015 Period: 21 months **Participating entities**: MATGAS 2000 AIE (Coordinator) (Spain), Air Products PLC. (UK); Ciaotech S.r.l (Italy); European Hydrogen Association (Federazione delle Associazioni Scientifiche e Tecniche) (Belgium); McPhy Energy SA (France); Politecnico di Milano (Italy); Solvay Specialty Polymers Italy S.P.A. (Italy); Technische Universiteit Eindhoven (The Netherland); Carburos Metálicos (3rd party of MATGAS 2000 AIE, Spain). **Budget:** 1.21 MM€
3. Hydrogen, Sustainability	**Title**: *HYDROGEN TRANSPORT IN EUROPEAN CITIES – HyTEC* **Coordinator:** Diana Raine, Air Products PLC, UK **Principal investigator (PI):** Lourdes F. Vega, MATGAS 2000 AIE **Funding entity:** European Union, Fuel Cells and Hydrogen Join Undertaking **Programme:** SP1-JTI-FCH.2010.1.1 Large-scale demonstration of road vehicles and refuelling infrastructure III. 7 (Demonstration project) **Start-End date:** 01/09/2011 – 31/8/2015 Period: 4 years **Participating entity/entities:** Air Products Plc. (Coordinator) (UK); BAA Ltd (UK); Centre of Excellence for Low Carbon and Fuel Cell Technologies (UK); City of Copenhagen (Denmark); Copenhagen Hydrogen Network (Denmark); Element Energy Ltd (UK); Fraunhofer IBP (Germany); Greater London Authority (UK); Heathrow Airport Limited (UK); Hydrogen Link (Denmark); Intelligent Energy Ltd (UK); HyER (Belgium); hySOLUTIONS GmbH (Germany); Ludwig-Bölkow-Systemtechnik GmbH (Germany); London Bus Services Ltd (TfL) (UK); LTI Vehicles (UK); MATGAS 2000 AIE (Spain); Hyndai Motors Europe (Germany). **Budget:** 29.5MM€
4. CO_2, Hydrogen	**Title**: *SOFC CCHP WITH POLYFUEL: OPERATION AND MAINTENANCE* – SOFCOM **Coordinator:** Massimo Santarelli, Politecnico di Torino, Italy **Principal investigator (PI):** Lourdes F. Vega, MATGAS 2000 AIE **Funding entity:** European Union, Fuel Cells and Hydrogen Join Undertaking **Programme:** SP1-JTI-FCH.2010.3.4: Proof-of-concept and validation of integrated fuel cell systems. (Demonstration project). **Start-End date**: 01/11/2011 – 30/4/2015 Period: 3.5 years **Participating entities**: Politecnico di Torino (Coordinator, Italy); Ecole Polytechnique Fédérale de Lausanne (France); Instytut Energetyki (Poland); MATGAS 2000 AIE (Spain); Italian National Research Council - Institute for Advanced Energy Technologies (Italy); Società Metropolitana Acque Torino (Italy); VTT Technical Research Centre of Finnland (Finland); Technical University of Munich (Germany); Topsoe Fuell Cell A/S; Università di Torino (Italy). **Budget:** 6.2 MM€
5. Hydrogen Fuel Cells	**Title:** *Long-life PEM-FCH&CHP systems at temperatura higher than 100ºC – LOLIPEM* **Coordinator:** Giuseppe Barbieri, National Research Council – Institute on Membrane Technology, CNR-ITM, Italy **Principal investigator (PI):** Lourdes F. Vega, MATGAS 2000 AIE **Funding entity:** European Union, Fuel Cells and Hydrogen Join Undertaking **Programme:** SP1-JTI-FCH.3.3 Degradation and life time fundamentals. **Start-End date**: 1/1/2010 – 31/12/2012. Period: 3years **Participating entities:** CNR-ITM (Italy), University of Roma (Italy), University of Provence (France), University of Saarbruecken (Germany), Edison (Italy), Fumatech (Germany), MATGAS 2000 AIE (Spain), Cracow University of Technology (Poland). **Budget:** 2.93 MM€
6. Hydrogen,	**Title**: *Nanostructured materials for solid-state hydrogen storage*

materials	**Coordinator**: Amelia Montone, ENEA, Italy **Principal Investigator (PI):** Lourdes F. Vega, MATGAS 2000 AIE (Spain) **Funding entity:** European Union **Programme:** Materials, Physical and NanoScience (MPNS) COST ACTION, MP 1103 **Start-End date:** 25/10/2011 – 24/10/2015. Period: 4 years **Number of researchers:** 165 researchers from 27 EU countries. **Budget:** 64 MM€
7. Catalysis and Separation, Modeling	**Title**: *Theoretical Chemistry and Computational Modelling- Joint Doctorate – TCCM* **Coordinator**: Manuel Yáñez, Universidad Autónoma de Madrid, Spain **Principal Investigator (PI)**: Lourdes F Vega, Alya Technology & Innovation, S.L. **Funding entity**: European Union **Programme**: H2020-MSCA-ITN-2014 (Topic: MSCA-ITN-2014-EJD) **Start-End date**: 1/1/2015- 31/12/2018. Period: 4 years. **Participating entities**: Univ. Autónoma de Madrid, Univ. de Barcelona, Univ. Basque Country, Univ. of Groningen, Katholieke Universiteit Leuven, Univ. Pierre et Marie Curie- Paris_VI, Univ. degli Studi di Perugia, Univ. of Pisa, Univ. of Porto, Univ. of Toulouse, Univ. of Valencia, Uni. of Vienna. Partner organisations: Atria Science SL, Barcelona Supercomputing Center, Biolitec Reserach GmbH, Consorzio Interuniversitario CINECA, GlaxoSmithKline Research and development LTD., Master-up, MATGAS 2000 AIE, PLC System, Scientific Computing and Modelling, Simune Atomistics SL, SmartLigs Bioinformática SL, Univ. of Stockholm, Fondation de Cooperation Scientique Campus Paris-Saclay, ZCAM - Universidad de Zaragoza. **Role in the project**: To host in our simulation group students, under my supervision, for long term stay to work in computational projects in an industrial environment, contributing to the advance of science and the final applications. In addition, I will collaborate with some training and education courses. **Budget:** 3.8 MM€
8. Hydrogen, Sustainability	**Title**: *Complex solid state reactions for energy efficient hydrogen storage - COSY* **Coordinator:** Klaus Taube, Helmholtz-Zentrum Geesthacht GmbH, Germany. GKSS-Forschungszentrum Geesthacht GmbH **Principal investigator (PI)**: Lourdes F. Vega, Carburos Metálicos/Air Products **Funding entity**: European Union **Programme**: FP6 - MOBILITY **Start-End date**: 1/11/2006 – 31/10/2010. Period: 4 years **Participating entities**: Helmholtz-Zentrum Geesthacht GmbH (Germany), University of Torino (Italy), Leibniz-Institute for Solid State and Materials Research Desdren (Germany), European Synchrotron Facility (France), University of Twente (The Netherlands), Materials Science Institute of Sevilla, CSIC (Spain), Karlsruhe Institute of Technology (Germany), Grenoble Institute of Technology (France), Vrije Universiteit, Amsterdam (The Netherlands), Universitat Autonoma de Barcelona (Spain), University of Oxford (United Kingdom), EMPA, Dübendorf (Switzerland), Laboratoire de Réactivité et de Chimie des Solides LRCS, Université de Picardie Jules Verne, Amiens, (France). Industrial Board: GfE Metalle und Materialien GmbH, Volkswagen AG, Fiat Centra Ricerche CRF, Carburos Metálicos, Shell Global Solutions International BV, Johnson Matthey Technology Centre. **Budget:** 2.5 MM€
9. CO_2 applications, Sustainability	**Title**: *Super Critical Carbon Dioxide Processing Tecnology for Biodegradable Polymers Targeting Medical Applications (PROTEC)* **Coordinator:** University of Birmingham, United Kingdom **Principal investigator (PI):** Lourdes F. Vega, Carburos Metálicos/Air Products **Funding entity:** European Union

	Programme: FP6-NMP **Start-End date**: 1/6/2005 - 30/11/2008. Period: 3,5 years **Participating entities:** U. Birmingham (UK), RAPRA (UK), TREXEL (Germany), U. Twente (The Netherland), U. Lodz (Poland), Carburos Metálicos (Spain) **Budget:** 2.2 MM€
10. CO_2 applications	**Title**: *Sustainable Surface Technology for Multifunctional Materials (SURFACE T)* **Coordinator:** Concepción Domingo, Instituto de Ciencia de Materiales de Barcelona, CSIC, Spain **Principal investigator (PI):** Lourdes F. Vega, MATGAS 2000 AIE **Funding entity:** European Union **Programme:** FP6 - NMP **Start-End date**: 1/06/2005-31/05/2008. Period: 3 years **Participating entities**: Consejo Superior de Investigaciones Cientificas (Spain), Bactimm BV (The Netherland), Centre Nationale de la Recherche Scientifique (France), European Medical Contract Manufacturing BV , Feyecon Development and Implementation BV (The Netherland), Fundacion Gaiker (Spain), Imperial College of Science, Technology and Medicine of London ((UK), J. Uriach y Cia SA (Spain), MATGAS 2000 A.I.E. (Spain), Midicamina Ltd (UK), Universidad de Barcelona (Spain). **Budget:** 3.6 MM€
11. Modeling & Experiments, Materials	**Title**: *Molecular simulations in Biosystems and Materials Science (SimBioMa)* **Geographical area:** European Union **Coordinators:** Berend Smit (contact person), CECAM/Ecole Normale Supérieure de Lyon, France; Peter Nielaba, Physics Department of University of Konstanz, , Germany **Principal investigator (PI):** LourdesF. Vega, Instituto de Ciencia de Materiales de Barcelona, ICMAB-CSIC **Funding entity:** European Union **Programme:** Research Training Network, European Science Foundation **Start-End date**: 2006-2011. Period: 5 years **Participant entities**: Universities and research institutions from different EU countries, more than 80 research groups involved, all related to molecular simulations. **Budget:** 1.1 MM€

OTHER RECENT R&D COMPETITIVE PROJECTS (BEFORE JOINING PI/KU, 2009-2015)

1. Modeling, Alternative solvents	**Title:** *Development of New Artificial Blood Substitutes using Fluorinated Ionic* Liquids **Participating entities:** MATGAS 2000 AIE, Universidade de Vigo, Universidade Nova de Lisboa **Funding entity:** TransBio SUDOE Interreg, Collaborative Projects, Biocat, EU **Period:** 2014 **Principal Investigators:** Ana Pereiro (UNL - Portugal), Manuel Martinez Pineiro (U. Vigo- Spain), Lourdes F. Vega (MATGAS-Spain) **N. of researchers**: 8
2. Modeling, Alternative solvents	**Title:** *Incorporating the correct scaling behavior into a molecular-based equation of state* **Participating entities**: MATGAS (Spain), University of Maryland (USA) **Funding entity:** Generalitat de Catalunya. Grants for short stays abroad. BE2012 **Period:** 2012 **Principal Investigators:** Felix Llovell (Spain), Lourdes F. Vega (Spain), Jan Sengers,

	Mikhael Anisimov (USA) **N. of researchers:** 5
3. Alternative energy, Nanotechnology, CO_2 utilization	**Title**: *Rational design of materials and processes for energy and nanotechnology applications.* **Funding entity**: Ministerio de Ciencia e Innovación. Proyecto CTQ2008-05370 **Principal Investigator**: Lourdes F. Vega, Instituto de Ciencia de Materiales de Barcelona, Consejo Superior de Investigaciones Científicas **Period**: 1/1/2009- 31/12/ 2011 **N. of researchers**: 11

COMPETITIVE GRANTS TO SUPPORT GROUPS OF EXCELLENCE IN R&D

(The research group was awarded "of excellence" in three consecutive evaluation calls)

2014 call	**Title**: *Applied Molecular Simulations (MOLSIMAP)* **Funding entity**: Generalitat de Catalunya. Grants to support consolidated research groups. **2014SGR-1582** **Participating entities**: MATGAS 2000 AIE, Instituto de Microelectrónica de Barcelona, Consejo Superior de Investigaciones Científicas (CNM-IMB-CSIC), Universitat de Barcelona, Universitat Rovira i Virgili. **Principal Investigator**: Lourdes F. Vega **Period**: 2014 - 2019 **N. of researchers**: 25
2009 call	**Title**: *Molecular Simulations (MOLSIM)* **Funding entity**: Generalitat de Catalunya. Grants to support consolidated research groups, **2009SGR-666** **Participating entities**: Grupo de Simulación Molecular, ICMAB- CSIC, MATGAS, URV **Period**: 2009- 2013 **Principal Investigator**: Lourdes F. Vega **N. of researchers**: 15
2005 call	**Title**: Molecular Simulation Group (MOLSIM) **Funding entity**: Generalitat de Catalunya. Grants to support consolidated research groups, **2005SGR-00288** **Participating entities**: Grupo de Simulación Molecular, ICMAB- CSIC **Period:** 1/2006- 1/ 2009 **Principal Investigator**: Lourdes F. Vega **N. of researchers**: 10

SOME INDUSTRIAL RELEVANT CONTRACTS (APPROXIMATE TITLES – CONFIDENTIAL WORK) – 2008-2015

Excluding the ones under execution previously mentioned

1. Materials, CO_2	**Title:** *Applications of $scCO_2$ to tailor-made materials* **Type of contract:** R&D **Company:** Henkel **Participants:** MATGAS, Henkel **Period:** 4/ 2014-5/2015 **Principal Investigator:** Lourdes F. Vega. **Participants:** 3 from MATGAS, 2 from Henkel
2. CO_2 utilization, Food, Sustainability	**Title:** *New applications of gases in agri-food* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products

	Participants: Carburos Metálicos, IRTA **Period:** 5/ 2014-8/2015 **Principal Investigators:** 2 from Air Products, 3 from IRTA
3. CO_2 utilization, Water Sustainability	**Title:** *pH neutralization of recreational waters with CO_2* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participants:** Carburos Metálicos, MATGAS, Universitat Autonoma de Barcelona **Period:** 5/ 2013-5/2014 **Principal Investigators:** Sonia Guri/Lourdes F. Vega (Air Products). Javie Lafuente (UAB)
4. CO_2 utilization, Nanotechnology	**Title:** *Design of new materials for CO_2 capture and utilization* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participants:** Carburos Metálicos, Instituto de Ciencia de Materiales de Barcelona - CSIC **Period:** 1/12-9/2014 **Principal Investigators:** Lourdes F. Vega (Air Products). Concepción Doming (ICMAB-CSIC)
5. CO_2 transportation	**Title**: *Research related to the CO_2-ducte of Compostilla* **Type of contract:** R&D **Company:** Endesa Generación **Participating entities:** Ciuden, MATGAS, Carburos Metálicos/Air Products **Period:** 12/ 2010-12/2011 **Principal Investigator:** Lourdes F. Vega
6. Clean energy, CO_2 utilization	**Title:** *Experimental characterization of injected supercritical CO_2 for CO_2 storage at Hontomin (Burgos)* **Type of contract:** R&D **Company:** Fundación Ciudad de la Energía - CIUDEN **Participating entities:** Ciuden, MATGAS, Amphos 21 **Period:** 06/ 2010-6/2011 **Principal Investigator:** Lourdes F. Vega, Joaquim Torres (MATGAS)
7. Clean energy, Bioenergy	**Title:** *Gasification of black liquors* **Type of contract**: R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos, MATGAS, Universidad de Zaragoza **Period:** 2009-2012 **Principal Investigators**: Lourdes F. Vega, Joaquim Torres (MATGAS), Jesús Arauzo (U. Zaragoza)
8. Hydrogen, Clean energy	**Title:** *Production of Hydrogen from Microbial Electrochemical Cells* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos. Universitat Autònoma de Barcelona **Period:** 2009-2012. **Principal Investigators:** Albert Guisasola (UAB), Lourdes F. Vega, Sonia Guri (CM/AP)
9. CO_2 utilization, Sustainability	**Title:** *Application of Life Cycle Analysis tools to selected new sustainable industrial applications of CO_2* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos. Centre Tecnologic Manresa (CTM) **Period:** 2008-2011. **Principal Investigators:** Miquel Rovira (CTM), Lourdes F. Vega, Roberta

	Pacciani (CM/AP)
10. CO_2 utilization, Sustainability, Food Preservation	**Title:** *Using Modified Atmosphere Packaging to preserve the selflife of food products* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos. Institut de Recerca i Tecnològica Agroalimentaria **Period:** 2008-2011. **Principal Investigators:** Jordi Riudavets (IRTA), Lourdes F. Vega, Sonia Guri (CM/AP)
11. CO_2 utilization, Water treatment	**Title:** *Valorisation of CO_2 for water treatment using membranes for direct osmosis* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos. CETAQUA, Agbar, Centre Tecnològic de Manresa **Period:** 2008-2011. **Principal Investigators:** Jordi Guimerà (CETAQUA), Lourdes F. Vega, Sonia Guri (CM/AP)
12. CO_2 utilization, photoreduction, nanotechnology	**Title:** *Development of new catalitic materials for CO_2 photoreduction using solar energy* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Carburos Metálicos. Fundació Bosch i Gimpera/IREC **Period:** 2008-2011 **Principal Investigators:** Joan Ramon Morante (IREC), Lourdes F. Vega, Oriol Ossó (CM/AP)
13. CO_2 utilization, photoreduction, nanotechnology	**Title:** *Photochemical reduction of CO_2 with solar light and nanoscopic semiconductors* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Universitat Autònoa de Barcelona (Quim), Carburos Metálicos. **Period:** 2008-2011 **Principal Investigators:** J.L Bourdelande (UAB), L.F. Vega, JO. Ossó (CM/AP)
14. Hydrogen, Nanotechnology	**Title:** *Hydrogen recovery from gas process streams* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** Universitat Autònoma de Barcelona (Física), Carburos Metálicos. **Period:** 2008-2011 **Principal Investigators:** Javier Rodriguez-Viejo (UAB), Lourdes F. Vega (CM/AP)
15. CO_2 utilization	**Title:** *Dry methane reforming assisted by microwave to obtain a syngas* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** INCAR-CSIC, Carburos Metálicos. **Period:** 2008-2011 **Principal Investigators:** Angel Menéndez, Ana Arenillas (CSIC), Lourdes F. Vega, Ignasi Salvadó (CM/AP)
16. CO_2 utilization, Nanotechnology	**Title:** *Using $SCCO_2$ for the preparation of high added value materials and functionalized surfaces (MAT superCO2)* **Type of contract:** R&D

	Company: Carburos Metálicos/Air Products **Participating entities:** ICMAB-CSIC, Carburos Metálicos. **Period:** 2008-2011 **Principal Investigators:** Concepción Domingo (CSIC), Lourdes F. Vega (CM/AP)
17. CO_2 capture and utilization	**Title:** *Evaluation and characterization of materials for CO_2 capture and CO_2 applications at high pressure* **Type of contract:** R&D **Company:** Carburos Metálicos/Air Products **Participating entities:** MATGAS, Carburos Metálicos. **Period:** 2008-2011 **Principal Investigators:** Joaquim Torres (MATGAS), Lourdes F. Vega (CM/AP)
18. Materials, CO_2 utilization	**Title:** *New materials and gases for food preservation using Modified Atmospheres Packaging* **Type of contract:** R&D **Company:** Nutreco **Participating entities:** MATGAS, Nutreco. **Period**: 2008-2011 **Principal Investigators:** Sonia Guri, Lourdes F. Vega (MATGAS), Alfredo Corujo (Nutreco)
19. CO_2 utilization, water treatment	**Title:** *Acidification of efluents from refrigeration circuits using CO_2* **Type of contract:** R&D **Company:** Iberdrola Generación **Participating entities:** MATGAS, Iberdrola. **Period:** 2008-2011 **Principal Investigators:** Sonia Guri, Lourdes F. Vega (MATGAS), Elías Rodriguez (Iberdrola)

ACADEMIC/TEACHING EXPERIENCE

SUMMARY OF ACADEMIC EXPERIENCE

Teaching experience as a regular faculty/lecturer in **five universities (USA, EU, ME)**: University of Sevilla (Spain), Cornell University (USA), Universitat Rovira I Virgili (Spain), Autonomous University of Barcelona (Spain), The Petroleum Institute, now part of Khalifa University of Science and Technology (UAE), Khalifa Unviersity of Science and Technology (UAE).

President of the PhD Program, Department of Chemical Engineering. URV, Tarragona, Spain. Define the strategy and initiate the International PhD program. 1996-99. 70+ students in 1999.

Deputy-Director (Associate Dean) for External & International Relations, *School of Chemical Engineering*, URV, Tarragona, Spain. 2001-03. Responsible for the bilateral relations between the School of Chemical Engineering and industrial companies; the School of Chemical Engineering and High Schools (to attract students and collaborate with the high schools), and the School of Chemical Engineering and International Universities and Research Organizations (Erasmus and other students/faculty exchange programs).

Regular Invited lecturer in some PhD and Master Programs in different universities in EU, USA and South America (Chile).

Senior Mentor of Junior Faculty: mentoring Assistant Professors joining the Chemical Engineering Department at the Petroleum Institute, part of Khalifa University of Science and Technology. Developing individual career paths with them and having regular meetings to follow up on execution, manage fustrations, and help in research funding opportunities and collaborations.

Courses taught: I have been teaching the following graduate and undergraduate courses, in 5 universities. In Chronological reverse order:

KHALIFA UNIVERSITY AND THE PETROLEUM INSTITUTE (PI), PART OF KHALIFA UNIVERSITY OF SCIENCE AND TECHNOLOGY, ABU DHABI, UNITED ARAB EMIRATES

GRADUATE STUDIES IN CHEMICAL ENGINEERING

1. **Molecular Thermodyanmics (CHEG750)** – Khalifa University, Department of Chemical Engineering, Elective course in the PhD program in Chemical Engineering. Spring 2019.
2. **Advanced Thermodynamics (CHEG630)** – Khalifa University, Department of Chemical Engineering, Mandatory course in the Master Studies in Chemical Engineering. Fall 2018
3. **Molecular Thermodynamics (CHEG589).** The Petroleum Institute. Department of Chemical Engineering. Special topic course in the Master Studies in Chemical Engineering. Fall 2016.
4. **Seminars (I and II) – Communication tools for graduate students (CHEG595 and 596).** The Petroleum Institute. Department of Chemical Engineering. Mandatory course in the Master and PhD Studies in Engineering. Fall 2016. Fall 2017

CHEMICAL ENGINEERING DEGREE

5. **Chemical Engineering Thermodynamics – CHEG332**. Department of Chemical Engineering. The Petroleum Institute (PI/KU). Mandatory course for Chemical Engineering Undergraduate Students (Junior year). Spring 2017, Spring 2018.
6. **Physical Chemistry– CHEG330**. Department of Chemical Engineering. The Petroleum Institute (PI/KU). Mandatory course for Chemical Engineering Undergraduate Students (Sophmore year). Fall 2017.
7. **Senior Design Project**. Advisor to two to three groups of students per semester (Fall 2016, Spring 2017, Fall 2017, Spring 2018, Fall 2018, Spring 2019)

UNIVERSITAT AUTONOMA DE BARCELONA (UAB), BARCELONA, SPAIN

PHD PROGRAM IN MATERIALS SCIENCE

8. **Simulation of materials at the atomic scale.** Chemistry Department. Fundamental course at the PhD program on Materials Science, (developer of the course, 2004-06). Invited Professor. Academic years 2004-05, 2005-06.

UNIVERSITAT ROVIRA I VIRGILI (URV), TARRAGONA, SPAIN

PHD STUDIES IN CHEMICAL AND PROCESS ENGINEERING

9. Advanced Thermodynamics. Universitat Rovira i Virgili. School of Chemical Engieering. Department of Chemical Engineering. Fundamental course in the PhD Studies in Chemical and Process Engineering. Invited Professor. Academic year 2003-04
Same course given as a Professor at the same University in the years 1995-96, 1996-97, 1997-98, 1998-99, 2000-01, 2001-02, 2002-03 (7 academic years).
10. Molecular and Process Thermodynamics. Universitat Rovira i Virgili. School of Chemical Engieering. Department of Chemical Engineering. Mandatory course in the PhD Studies in Chemical Engineering. Year 1999-2000.
11. Molecular Simulation of Materials. Universitat Rovira i Virgili. School of Chemical Engineering. Department of Chemical Engineering. Optional course in the PhD Studies in Chemical Engineering. Year 1999-2000.

CHEMICAL ENGINEERING DEGREE

12. Thermodynamics for Chemical Engineers. School of Chemical Engineering. Department of Chemical Engineering. Mandatory course for Chemical Engineering Students (3r semester of their studies, 2nd academic year). Years 1996-97, 1997-98, 1998-99,1999-2000, 2000-01, 2001-02 (6 academic years).
13. Principles of Polymer Systems. School of Chemical Engineering. Department of Chemical Engineering. Optional course given at the 5th semester of the Chemical Engineering Studies (3rd academic year). Years 1997-98, 1999-00, 2000-01, 2001-02, 2002-03 (5 academic years).
14. Communication tools for Engineers. School of Chemical Engieering. Department of Chemical Engineering. Optional course for Chemical Engineering Students (3r semester of their studies, 2nd academic year). Years 1999-2000, 2000-01, 2001-02 (3 academic years).
15. Industrial Practicum. School of Chemical Engineering. Department of Chemical Engineering. Mandatory course for Chemical Engineering Students (7th semester of their studies, 5th academic year). Years 1999-2000, 2000-01, 2001-02, 2002-03 (4 academic years).
16. Engineering Final Project. School of Chemical Engieering. Department of Chemical Engineering. Mandatory course for Chemical Engineering Students (last semester of their 5th years degree). Years 1998-99, 1999-2000 (2 academic years).

INDUSTRIAL TECHNICAL ENGINEERING, FOCUS: INDUSTRIAL CHEMISTRY

17. Reactor Design. School of Chemical Engineering. Department of Chemical Engineering. Mandatory course for Industrial Technical Engineering Students (3r semester, 2nd academic year). Year 1994-1995.
18. Communication tools for Engineers. School of Chemical Engieering. Department of Chemical Engineering. Optional course for Industrial Engineering Students (5th semester, 3rd academic year). Cursos 1999-2000, 2000-01 (2 academic years).
19. Principles of Polymer Systems (Polymers Science and Technology). School of Chemical Engieering. Department of Chemical Engineering. Optional course for the Industrial Engineering Students (6th semester, 3rd academic year). Years 1995-96, 1996-97 (2 academic years).

CORNELL UNIVERSITY, ITHACA, NY, USA

PH. D. COURSES

20. Theory of Molecular Liquids, Chem. Eng 772. Cornell University (USA). School of Chemical Engineering. PhD Program in Chemical Engineering. Co-lecturer with Dr. Keith E. Gubbins. Responsible for the part of polymer fluids and solutions. Fall 1994.

21. Atomistic Simulation of Materials, Chem. Eng. 774. Cornell University (USA), School of Chemical Engineering. PhD Program in Chemical Engineering. Co-lecturer with Dr. Athanassios Z. Panagiotopoulos (responsable of 1/3 of the total course). Spring 1993.

CHEMICAL ENGINEERING DEGREE

22. Thermodynamics for Chemical Engineers, Chem. Eng. 313. Cornell University (USA). School of Chemical Engineering. Junior Year. Co-teacher with Professors Keith E. Gubbins and Paulette Clancy. Fall 1994.

UNIVERSIDAD DE SEVILLA, SEVILLA, SPAIN

BACHELOR OF SCIENCE IN PHYSICS

23. Statitistical Mechanics. Universidad de Sevilla. Department of Atomic, Molecular and Nuclear Physics. Mandatory annual course (2 semesters) for the degree of Bsc in Physics, 3rd academic year. Year 1989-1990.

24. Collaboration with different teaching activities in the area of Theoretical Physics at the Department of Atomic, Molecular and Nuclear Physics. Year 1989-1990.

INDUSTRIAL TECHNICAL ENGINEERING

25. General Physics. Universidad de Sevilla. Polytechnical School. Department of Applied Physics. Mandatory annual course (2 semesters) in the 1st academic year of the Industrial Technical Engineering Studies. Theoretical and Laboratory lectures. Year 1988-1989.

ADDITIONAL RECENT COURSES IN WHICH I PARTICIPATE AS INVITED LECTURER

26. From University to Business (from U2B). Autonomous University of Barcelona, Spain. General graduate course. Years 2013-14, 2014-15. Lectures on how to apply research and scientific knowledge in industry.

27. Master on R&D Management. University of Zaragoza, School of Engineering. Spain. Inaugural lecture on Open Innovation. Years 2016-17, 2017-18, 2018-2019.